The problem

IS India facing a crisis of learning? Are we facing a teaching crisis? Is our education system in deep crisis? These questions elicit very different responses from different stakeholders and from experts favouring different theoretical frameworks. However, they all share a common apprehension – when they complete 8 or 10 or 12 years of schooling, many of our children are not equipped to negotiate the world they live in with confidence and with the requisite knowledge and skills to help them deal with the world from a position of strength. Whether it is formal skills to read and write, cognitive skills, technological skills or other higher order thinking skills – parents and children admit that the schools and colleges they attended did not add much value.

Equally, most observers also agree that the 'social capital' that children bring into the school is an important predictor of 'success', meaning that children who have educated parents, have access to books and other reading materials, have greater exposure to creative arts, media and live in resource-rich environments, seem to gain a lot more from the educational process as compared to those who come from resource-poor environments. Conversely, children from socially and economically disadvantaged communities who face blatant and subtle forms of discrimination inside the school, from their teachers and fellow students, leave school with low self-esteem and confidence and very little 'learning'.

Girls carry an additional disadvantage as they move higher in the academic ladder – they do not get the subject of their choice and in many states (especially in some northern and western states) girl's secondary schools do not offer science, mathematics or commerce. Sexual harassment on the way to school and in the school remains a huge issue. Similarly, children in tribal areas and from the most disadvantaged tribal communities not only experience discrimination but have far poorer access to schools beyond the elementary level. Yes, we have been facing a crisis of confidence in our education system – and this is not new.

India has a rich tradition of education and over time, different kinds of systems have coexisted. However, for the last two-hundred years the formal school system adopted during the colonial period has gradually become the dominant model. Yet, at least till recently, there was some space in India for alternative models of formal education with a rich pool of schools that tried out different approaches and pedagogies. With the coming of the RTE Act the space for alternative approaches and models seems to have shrunk. Equally significant is that over the last few decades there is a growing apprehension about the quality of people who choose to enter the teaching profession and the quality of teacher education institutions. The examination system has also become more memory-centric and there is little scope to assess anything else. Most professional courses have entrance examinations because they are wary of going by the results children obtain in board examinations. Thousands of coaching centres and tuition classes have cropped up across the country – emerging as parallel educational institutions, one that is not regulated.

In the last few decades concerns have also been raised on the systematic stratification of schools according to social, economic and locational advantage/ disadvantage. Who you are, where you live, how much your family earns and your social identity determines the kind of school (private and government) you go to and the kind of education you receive. This is important because the kind of school a child goes to determines/influences her learning.

It is in this scenario that we need to think about the following interrelated issues. *One*, what enables a child to learn at her own pace, explore her potential and progressively benefit from schooling? *Two*, what can empower our teachers to focus on their students and facilitate their learning? *Three*, what can the administration do to create an environment for meaningful teaching and learning, and not be content with counting numbers of children enrolled, numbers of classrooms, mid-day meals distributed or the number of teachers trained. And *four*, what can the larger education community (including civil society organizations engaged in education) do to support the children, the school and the teachers. These questions would invariably lead us to look at both systemic issues that have stifled our education system but also the widespread hierarchies and discrimination that is embedded in the school system. Intriguingly, a vocal section of the education community seems to be attracted to large-scale testing as a tool to 'reform' the system. What then is the global evidence on the contribution of large-scale testing to educational reform?

A number articles and books published in the last few years (notably Daniel Koretz's *The Testing Charade*[1]) argue that standardized tests and large-scale assessments that have become very popular across the world, do not help in improving learning levels. The belief that quantitative metadata helps decision making is based on very weak evidence. Speaker after speaker in a recent (April 2018, New Delhi) World Bank consultation on large-scale assessment argued that large-scale assessments have not resulted in any substantial reform. Countries that did reform the system started with the teachers, the school, the examination system and the overall school environment.

Diane Ravitch[2] argues that for 'the last 16 years, American education has been trapped, stifled, strangled by standardized testing… the pressure to raise test scores were *inflated by test preparation*… some administrators gamed the system by excluding low scoring students from the tested population…' Similarly, Koretz argues that the reform movement failed miserably because of its devotion to high stake testing 'as the infallible measure of education quality'. He argues that the obsession 'prepares students to take tests, but it does not prepare them to apply what they have learned to real life situations'.

For over thirteen years ASER (2005 to 2015, and most recent 2017) have highlighted the poor learning levels among our children. The most recent survey of 14 to 18-year-old youth has thrown up extremely disturbing results – not only on formal learning but also on the low levels of knowledge and confidence among our youth. A recent working paper by Research on Improving Systems in Education (RISE) compares school leaving exams in India, Pakistan, Uganda and Nigeria and concludes that even compared to Uganda and Nigeria, the Indian (and Pakistani) exams were extremely rote-based.[3]

Those involved in the National Assessment Survey (conducted by NCERT) tell us that the learning situation is not uniformly bleak and that there are wide regional variations across India, as the voluminous district-wise reports surveyed over 100,000 students in government schools in over 700 districts shows. Reports in the media suggest that on average, a class VIII student could barely answer 40% of the questions in maths, science and social studies. The national average score for language was a little better at about 56%. In class III, students averaged between 63% and 67% marks in environmental science, language and maths. In class V, average scores fell by about 10 percentage points to 53-58%, and in class VIII, the fall was even sharper, although language scores dipped only a little. No other subject was common between these classes.

Among regions in India, the South did better in the early years of schooling. For example, in class III, students in the South scored more than those in the North and the East, and were behind those in the West only in language. In class V, too, they had higher average scores than the rest, but in class VIII, students from the West outperformed all other zones. The South remained ahead of the North, but scored less than the East in science and social studies. Rural students scored higher than those in cities. Also, in classes V and VIII, OBC (Other Backward Class) students outscored the general category. At all levels, average scores were lowest for ST

(Scheduled Tribe) students while SC (Scheduled Caste) students scored a tad higher.[4]

In what way have these tests results influenced the education scenario in India? Have they been used as diagnostic tools to help the teacher in the classroom? Have they been used by our curriculum planners and administrators to understand what is happening inside the school? Has it nudged our administrators to not push to finish the course/textbook and focus instead on what and how much the children are learning? The big question is – have these national tests led to any concrete reform? While the jury is still out on this question, the only tangible 'impact' (so to speak) seems to be greater public concern about the levels of learning of our children – and that is definitely a good thing.

Some states and NGOs believe that short-term mission-mode projects for basic reading/accelerated learning would somehow restore the balance. Some states like Tamil Nadu decided to intensify school monitoring by district and sub-district level administrators. Many state government officials and politicians believe that if the no-detention policy is withdrawn children will start learning. The current dispensation in the central government also seems to believe that scrapping the no detention policy will miraculously change the situation. Notwithstanding compelling global evidence (especially from Finland, Singapore, Poland to name a few) and the mounting evidence that shows no positive correlation between large-scale testing and educational reform, we in India continue to look for magic bullets that would dramatically alter the scenario.

The challenges we face are only in part due to a lack of political will; it is because we have refused to address systemic issues. Our short-term and immediate result oriented approach has prevented long-term planning and strategizing to make the education system the engine of human development. Among other reasons, we have not invested in creating a professional discipline and expertise around providing quality education at scale. Generalist administrators are in charge and many of them overturn what their predecessor did and try to do something new. We all know that the sector needs patient effort rather than short-term measures aimed at showing quick results within political or administrative terms in office. As a consequence of this tendency, we have focused on tangibles like buildings and hardware – and this too has been riddled with corruption and inefficiency.

For a long time now, we have ignored our teachers, basically made almost no effort to produce high quality and knowledgeable teachers. We have neglected teacher support systems and, most importantly, we have treated teachers as the last rung in a hierarchical bureaucracy. Our monitoring systems have focused on input data and very little on both tangible learning outcomes as well as the intangibles like overall development of our children. Rote-based exams that have changed very little over the last six decades, have further eroded our system.

On the positive side, the commitment and efforts to address educational issues remain extremely high across all sections of society – from the very poor to the very rich. There is a growing awareness about the importance of education and people from all walks of life are trying to get the best they can for their children. Unfortunately, what is available to parents are tuition shops and coaching centres. Many more non-governmental players are active in small-scale efforts to enhance learning in schools and working with teachers. Some are trying to work on a larger-scale across several states. There are also those who are lobbying for greater privatization and more frequent testing. There are others who believe that technology has all the answers and are pushing for a teacher-proof curriculum.

The education landscape is crowded with competing interests. In the middle of this are teachers – the only people who can actually make a concrete difference on the ground. They unfortunately have little voice in this debate and continue to be vilified and blamed. Teachers teach as they themselves were taught, unless they are provided with opportunities to analyse and question their own experience and thereby construct a different conception of what classroom processes should aim to achieve. In the last twenty years we have spoken to hundreds of teachers – almost all of them say that they are expected to ‘follow orders’, ‘cover the syllabus’, ‘fill out formats’, and so on. It is hard to think of a more damning indictment of the education system than this: teachers do not even conceive of their work in terms of creating an environment where all children can learn. Changing this would involve keeping our ears close to the ground, listening and acting in such a way that we take everyone along.

This issue of *Seminar* will hopefully enable us to appreciate the complex landscape of learning, and more, help us ask the ‘right’ questions to effect meaningful change in our learning environment.

VIMALA RAMACHANDRAN
SRIDHAR RAJAGOPALAN

Footnotes:

1. Daniel Koretz, *The Testing Charade: Pretending to Make Schools Better*. University of Chicago Press, Chicago, 2017.

2. Diane Ravitch, ‘Settling the Scores’, *The New Republic*, 27 December 2017.

3. Newman Burdett, Review of High Stakes Examination Instruments in Primary and Secondary School in Developing Countries. RISE Working Paper 17/018, 2017. https://www.riseprogramme.org/sites/www.rise programme.org/files/publications/RISE_WP-018_Burdett.pdf

4. *The Times of India*, 24 February 2018.

Diagnostic trouble

KRISHNA KUMAR

FOR an illness to get cured, they say, it is important to identify it correctly before prescribing, let alone administering, a medicine. I recommend this traditional wisdom for application in the field of education. A lot of well intentioned, sincere people in our country and abroad are deeply unhappy with India's system of school education. They want to 'fix' it – a term widely used these days to indicate urgency in finding a solution, but also indicating some impatience with the demands that understanding a problem makes.

According to many among education fixers, the problem India faces in its schools, especially state run schools, is that of learning. They feel that there is a crisis of learning. This feeling is backed by evidence of the kind widely favoured today. It is derived from the results of tests conducted through vast surveys. The results are in the form of scores achieved by a sample consisting of lakhs of children at different stages of elementary education. The size of the sample, its spread across the country and the frequency at which the tests have been administered are all impressive. Year after year, the voluminous data presented in the shape of the Annual Statistics of Education Report (ASER) show that children are doing very poorly in basic areas like literacy and numeracy. This finding has served as the basis of the view that there is a nationwide crisis of learning.

The main link between the scores and the conclusion is the expectation that children's growth in learning follows a normal pattern, meaning that if a child studying in Grade 4 cannot or does not comfortably read a text considered appropriate for Grade 2, then we should be worried about this child's learning. Prima facie, this reason looks sensible, especially if we are looking at an individual case. We should indeed worry about the child studying in Grade 4 or 5 who cannot read a Grade 2-level text, but the nature of our worry is no small matter. Any number of reasons can cause the situation we are describing, and our worry ought to make us curious about these reasons. Now if you multiply this single case by a million times, the problem of interpreting test scores becomes highly complex.

Many analysts might even say that scores acquired through large-scale tests have little use. If a vast number of children are found to be way behind the norm set for learning levels, any number of things might constitute the reason of this lag, starting with the quality of the test itself. Other reasons would include the conditions under which the test was administered. When one considers India's linguistic diversity, the problem of validity and comparability of children's scores in different regions becomes formidable. Add to this the various dimensions of social inequality that our system of education copes with.

Looking at the vast array of factors shaping children's performance in a country as large and diverse as India, one begins to wonder about the worth of the test-based survey method to assess the health of the education system. At any rate, this train of thought would lead one to question the clarity of the conclusion that India is faced with a learning crisis.

ASER surveys have convinced many people about the existence of a learning crisis. Armed with the vast statistical treasure that ASER has provided for well over a decade now, they recommend a slew of remedies. Many of these remedies are already being administered. These remedies offer further revelations about the perception of the learning crisis shared by three major components of the educational decision-making world: civil servants, officials of international donor agencies, political leaders and NGOs (also known as 'activist' and civil society groups). The remedies they have introduced are meant to increase two main values: accountability and efficiency.

Typically, ASER data, showing low learning rates, are read and presented along with teacher absenteeism data. These too have been collected with the help of one-off surveys of schools. The widely shared findings of such surveys have served to link the learning scores of ASER with truancy among teachers. This linkage has, in turn, served as the basis for the decision in many states to initiate different kinds of accountability regimes for teachers, ranging from biometric attendance, to mandatory wearing of uniform in high visibility colours, such as pink and black. The former is meant to ensure that teachers come on time and stay through the school day; the latter ensures that a truant teacher is spotted when he or she is physically away from school. Another remedy to ensure teacher accountability is involvement of parents in maintaining a vigil.

In addition to these measures, learning crisis diagnosticians suggested providing external support for improving the capacity and pedagogic engagement of regular teachers. This idea has taken various forms in different regions. A highly educated volunteer from other fields spending a few months at a school is one model; a low-paid volunteer from the school's neighbourhood is another. Another idea is to tweak the curriculum. In many cases, it has been suggested that the curriculum should be dumbed down in order to make it more realistic. A third strategy is to technologize learning, by introducing computers in primary classes and using distance education methods to upgrade the teacher's knowledge and skills.

Use of technology, such as CCTV cameras, to enable administrators and parents to keep an eye on what is going on in the classroom is also a favoured strategy currently being applied in many states to make teachers more accountable. In some states, government schoolteachers are required to wear a uniform, such as a pink sari and black blouse, or a jacket with the sentence '*Mein rashtra sevak hoon*' (I am a servant of the nation). These dresses are believed to make visible the teacher's presence in spaces other than the school during working hours. All such measures are based on the assumption that the so-called learning crisis has been caused by non-performing teachers.

Turning to an alternative diagnosis now, let us start by changing the name of the crisis. Instead of calling it 'a learning crisis', let us try calling it a 'crisis of teaching'. This is not a mere labelling issue as some might retort. By spotting the crisis in teaching rather than in learning, one is changing the nature of the debate over what is wrong and how best it can be put on the course of getting corrected.

By shifting our focus from learning to teaching, we get a different view of the educational scenario. To begin with, we recognize the teachers as major players. We notice that they are unhappy and angry across the country, particularly in northern India where the crisis afflicting the system is graver. We also notice that a vast number of teachers are not there, not just on the day we are doing a survey, but permanently so, and the reason is that they have not been recruited and appointed. The exact number of teaching posts lying vacant for years is hard to guess, but according to commonly used national aggregates, it is more than a million.

The meaning of such a high figure of vacancies takes an effort to grasp. How does one visualize the absence of a million plus teachers? It may be useful to try making sense of a small territory like Delhi where about 15,000 teaching positions lie vacant at any given time. How do schools cope with this shortage? Let us imagine a school which has seven posts to cover the different subjects, but only four are actually available, not because the remaining three are absent, but because they have not been appointed. If one is the headmaster of such a school, one would have to cope with the shortage by merging two or more classes or letting some periods remain teacherless. Both steps will have consequences for learning.

Cumulative growth of shortage of teachers due to non-recruitment has been met by hiring of temporary teachers who lead vulnerable lives. The term 'vulnerable' covers a range of categories used in different states. In the beginning, they were seen and called by the name 'para teachers'. Criticism and resistance led to the use of euphemistic terms like '*guruji*', '*shikshamitra*' and '*vidyamitra*'. The new teachers appointed under the District Primary Education Programme (DPEP) and Sarva Shiksha Abhiyan (SSA) to cope with expanded enrolment were mostly contractual or temporary. Their emoluments were significantly lower than what their permanent colleagues working in the same schools were receiving.

The number of such teachers kept growing over the 1990s, while the number of permanent teachers was declining due to retirement. In Madhya Pradesh, the permanent cadre was officially labelled as a 'dying' cadre, indicating the policy that this cadre will not be replenished. Now, after more than two decades of this transition, the state has substantially brought down the level of emoluments a person wanting to be a teacher can expect. Despite such manoeuvres, shortage of teachers continues to run into tens of thousands. The greater these numbers, the more difficult to manage gets the task of recruitment, even at lower emoluments.

In the sphere of teacher training, vast changes have hit the system. Applicants for teaching positions have degrees from certified institutions, but the overwhelming majority of these institutions are known to be academically hollow, and a major proportion are simply fraudulent. However, they are affiliated to the recognized universities, so the degrees their students append to their applications cannot be questioned.

To cope with this problem, the government has now instituted an eligibility test. Only a very small number of applicants have proved their eligibility by passing this test, and thus the pool available to meet the shortages of teachers in different subjects has shrunk, exacerbating the problem of non-recruitment and persistence of shortages. Such a complex landscape can be expected to have plenty of scope for litigation. In many states, people serving for several years on a temporary basis are able to leverage the courts to seek a stay on fresh recruitment for permanent vacancies. That is yet another factor responsible for making shortage of teaching staff in schools a normal reality.

Isn't there a link between a decline of teachers' status as employees and children's academic performance? Such a link seems obvious. Yet, there has been little criticism of the various decisions taken over the last three decades in several states in the context of teacher recruitment and emoluments. ASER ignores the condition of teachers; so do lobbyists who use ASER data to debunk government schooling.

On the contrary, they argue that teachers are responsible for the poor academic performance of children. Circular logic is applied to suggest that teachers who teach poorly or don't teach at all ought not be paid well. An impression has been created that government schoolteachers are overpaid. This view regards the growing tendency among the poorer sections of society to send children to low-fee charging private schools, where teachers work for paltry sums, as the basis to say that government schoolteachers get emoluments they don't deserve.

Embedded in this argument is a tacit approval and appreciation of policies like contractual appointments and lowering of the salary and status of teachers. Social acceptance of these trends has also grown on account of parental preference for English-medium education, no matter how poor its quality may be. Governments have responded by introducing English from Grade I itself, but that decision makes little difference to the general perception that state run schools are worthless as their standard is low and teachers don't care for children.

The decision to introduce English from the start of primary education does not include recruitment of teachers trained to teach English. In the wilderness of India's education system, all decisions now constitute a response to a crisis that no one has time to contemplate.

In the context of training too, any systemic debate on problems and choices has become impossible to mount. The view that training makes no difference has gained currency. It can be seen as one of the many outcomes of the tendency to encourage fully private institutions in teacher training and the government's inability to enforce sensible regulatory mechanisms to ensure standards.

The National Council of Teacher Education (NCTE), which is supposed to play this regulatory role, has failed to secure its own institutional health. A medical report on its awful condition exists in the shape of two volumes produced by a commission appointed by the Supreme Court under the headship of the late Justice J.S. Verma. These volumes eloquently bemoan, analyse and recommend. Some action did get initiated as a result of the court's pressure, but it was not sustained.

It is hard to notice any signs of institutional recovery in teacher training at the moment. If one believes that training equips teachers to play their pedagogic role, one has no reason to hope for an improvement at present. Given this impasse, the outcry against the learning crisis will continue to ignore the crisis in teacher training just as it has ignored the crisis in teacher recruitment and working conditions.

All the things I have described so far fit nicely into the general picture that critics of economic liberalization draw. Indeed, some readers of this article might wonder why I have not used the term 'neo-liberal' so far to critique the various changes I have touched upon. The term 'neo-liberal' would have served as a convenient signpost, letting at least some readers leave the rest of the article unread. So sharp is the division between supporters of economic reforms and their critics that the two sides do not overlap on any matter, including the various things discussed so far in this article about learning, teaching, training, and so on.

Those who have labelled the present situation as a learning crisis apparently think that there is no point fighting the imperatives of liberalization that have already set in. The system must cope with them, they seem to believe. This position implies an acceptance of two hypotheses: one, the diminution of teaching as a profession was part of the liberalization package; two, there is no going back.

Both these hypotheses can be debated. While it is true that liberalization and related policies required downsizing of the state by cutting government jobs in sectors like teaching, one can still hold the view that the various state governments, especially in the North, went much too far in this respect. Both reducing public spending by contractualizing and other means to make services cheaper to provide as also simultaneously encouraging the market to provide services to compete with the state, were indeed part of the global package of neo-liberal advice.

The response to this advice received differs from country to country. Within India too, we see several different types of response measures. Had the ideological component of neo-liberal advice been compulsive, Tamil Nadu's behaviour should have been the same as that of Madhya Pradesh and Bihar. The southern states, however, have navigated through the winds of liberalization without letting their essential systemic health getting affected. Schools do not look as damaged in the South as they do in the Vindhya-Gangetic belt, Rajasthan and further North. The question whether the North can recover from the injuries it has inflicted on its system does not necessarily mean a return to the 1980s.

Several good things happened under SSA, and they need to be conserved by institutionalization. The late Anil Bordia showed just how the new, temporary structures created under DPEP and SSA can be harmonized with the old, steel-like directorate-centred system. Similarly, the Verma Commission report shows how sanctity of teacher education can be restored. And this report is equally relevant to the South as it is to the North.

Let us now turn to the unfamiliar crux of the problem of school teaching: its dependence on the quality of higher education. Teachers at all levels, including the nursery, are ultimately the products of higher education. The reasoning underlying this somewhat strange formulation is not all that strange. Teachers serving the higher secondary level, i.e. Classes XI and XII, require a Master's degree. Those serving between Classes VI to X require a Bachelor's degree. They get their basic subject knowledge in colleges and universities. No matter how hard a training institution tries, it cannot compensate for a poor grasp of the subject. This is why the problem of learning at the school level is a product of systemic negligence and decay so manifest in undergraduate colleges across the country.

For the nursery and primary level, eligibility to become a teacher does not legally require an undergraduate degree (which, ideally, it should). This does not mean that poor quality undergraduate education does not affect nursery or primary level teaching. The cascade effect of bad teaching at higher levels goes down all the way.

The opposite argument, though more popular, is flawed. This is the argument that college teaching suffers because students coming to college have poor grasp and no interest. There are two reasons why this argument is flawed: first, because it blames the student, and second, because it ignores a key function of higher education on which comprehensive school improvement depends. Blaming the young is futile because they depend for their education on adults; so, no matter which age group we blame, the argument will remain a fallacy.

The second reason is more obvious. One of the key functions of higher education is to produce knowledge and to put students in touch with whatever is current or latest. A nursery or primary teacher cannot on her own choose to be guided by the latest psychological wisdom about children. She will be guided by whatever has been taught at the college or still higher level, not necessarily to her directly, for she may not have gone to college, but to whosoever taught her at a training institution.

This brief discussion brings out the peculiar position of teacher training as a form of knowledge that straddles school and higher education. In functional terms, good teacher training depends on cooperation between higher and school education. During their training, teachers must practice what they are learning, and for practice, they need positively inclined, pleasant school settings where the older teachers and principals appreciate the visiting young trainee teacher.

This is where our system fails almost totally. The cussedness with which trainee teachers are received by government school principals is matched by reluctance of the education officers in the directorate to give permission to training colleges to send their students for teaching practice. In private schools too, a principal who is happy to receive young trainees for a few weeks or months is rare. Some of the permanent teachers in private schools take the arrival of the trainee as a source of temporary relief from their demanding life; others view it as a time to criticize and harangue.

I hope I have explained why I prefer to call the present crisis of education a teaching crisis rather than a learning crisis. However, beyond a point, it does not matter what we call it. What we should worry about is the reality that no government – neither at the Centre or in any of the states – is particularly bothered and the burgeoning private sector is also not eager to raise an organized alarm. Complacency can only facilitate our continued fall.

Reforming education systems

YAMINI AIYAR

BACK in 2010, I participated in a meeting on education reforms with some of Bihar government's senior education officials and reformist politicians. In the course of the discussion, a senior official made an important observation. 'The government can', he said (and I paraphrase here) 'change how it works in short bursts. We can reform teaching practices and innovate in summer camps and mission mode programmes but when it comes to mainstreaming changes in our day-to-day work, we fail.'

A few years later, this same sentiment was echoed by another Indian Administrative Services (IAS) officer when he described India as the 'burial ground of pilots'. Both officers were pointing to an important puzzle that education reform efforts in India must confront. Government education systems are capable of reforms when the change is small-scale, mission oriented and time bound. But when it comes to embedding reforms in its everyday functioning and scaling up, the system fails.

Intrigued by this observation my colleagues and I have spent the last few years studying the education system at the frontlines in an attempt to unpack this puzzle. Our focus has been on understanding the everyday practices of local education bureaucrats, the decision making systems within which they function and the organizational culture and norms that this fosters. We then used this lens to understand how reforms are interpreted, implemented and institutionalized on the ground. To better understand reform failures we were particularly interested in studying how and why reforms are resisted, subverted and ultimately distorted. And of course, what worked.

The first step in our research was a study of how frontline actors (teachers and bureaucrats) interpret their role within the education system. When we asked our interviewees to describe their roles, the most frequently cited self-description was of being powerless cogs in a machine or 'post offices' used simply for doing the bidding of higher authorities and ferrying messages between the top and bottom of the education chain.

As one Block Education Officer (BEO) said of one of his critical tasks, managing teacher training – 'I spend much of my time ensuring everyone knows when the teacher or staff trainings are, ensuring all teachers and my own staff arrive on time and their TA/DA is paid.' When asked whether he engages with issues related to pedagogical content or teacher training modules, he offers an answer echoed in many of our interviews: 'I don't think that's really my job.' One reason for this limited understanding of their role is the perception amongst officers that their voices were rarely heard. '*Humari awaz kaun sunta hai*', was a common response we got when we asked our interviewees whether they had any influence over the decisions regarding implementation.

This sense of powerlessness runs so deep that many intervieweed referred to the '*Sarkar*' as something outside of themselves and over which they have no control. '*Agar sarkar chaahti hai to bahut kuch kar sakti hai*', was a frequent response when we asked our interviewees what could be done to improve education. But very few were willing to discuss their potential agency in making this change. This was most visible when we spoke to teachers and school administrators about the challenge of learning in government schools. In 2014, my co-authors Ambrish

Dongre, Vincy Davis and I interviewed 110 frontline education administrators in Bihar to capture their perspectives on the constraints to children's learning in elementary schools. Our interviewees viewed the challenge of learning primarily as a consequence of circumstances outside their control. These included poor policy, poor administration, lack of interest from parents from above, amongst others. But the classroom, the role of teachers and their relationship with students and the interviewees own agency never entered the discussion. Expectedly the solutions too lay outside their domain of influence. After all, if the government wants, it can do anything!

Research on India's frontline education system has quite conclusively shown the frontline to be powerful, unruly and often corrupt where absenteeism, low effort and apathy are the norm.[1] This contrasts sharply with the narrative of powerlessness that the frontline actors we interviewed have crafted for themselves. A narrative that runs so deep that most, as the quote above so effectively demonstrates, consider the state as something outside of the very actors that embody the state. It is easy to dismiss this narrative of powerlessness as a deliberate strategy deployed by an apathetic and corrupt bureaucracy. Regardless, it is important to understand how such an atmosphere is produced and sustained and what impact this has on how implementing agents understand their roles and make meaning of their jobs.

In his analysis of India's education system, Lant Pritchett hypothesizes that the reason why India (and many other developing countries) demonstrate low progress on basic learning outcomes, is because the education system is designed to cohere only around the goal of schooling inputs (enrolment, access, infrastructure) rather than learning.[2] Our exploration into the organizational culture of the frontline education system suggests that this input focused system design has served to produce and entrench the narrative of powerlessness or what my co-author Shrayana Bhattacharya and I define as the post office syndrome.

Arguably, a system designed to produce inputs necessitates a post office culture where implementing agents follow orders, supply information and comply with orders received and monitored from the top. Hierarchy and rules are the hallmark of such a system and this in turn shapes officers account of their jobs and how they go about their daily routines. On two separate occasions (2013 with Block Education Officers – BEOs – and 2014/2015 with Cluster Resource Centre Co-ordinators or CRCCs, administrators charged with providing pedagogical support to teachers), we conducted time-use surveys to discover the extent to which input focused goals and associated orders shape the daily lives and understandings of their role as education administrators.

We usually found that BEOs start their day with phone calls from their district bosses informing them of new input related 'government orders' received and the tasks they have to perform. As a result, each day is spent executing unplanned tasks rather than fulfilling the tasks they were hired for. During our study, Bihar's BEOs were busy implementing orders to organize camps for uniform and scholarship distribution. In Himachal Pradesh, BEOs were busy managing exams while Andhra Pradesh's BEOs were implementing teacher recruitment orders. During this time, none of the officers found any time to respond to reports received or needs expressed by those who visited their office. In fact, it was common for headmasters (HMs) and village elders who visited the block officers to raise concerns about their schools to be asked to wait while BEOs completed their district specified tasks.

In its functioning, the entire block office appeared to be geared toward responding to orders rather than responding to the needs of the school. In fact 'learning' related activities found almost no place in the daily activities of the block office for the time period of this study. This lack of focus on learning was even starker in our study of CRCCs. On an average visit to a school, CRCCs surveyed spent a mere 10-20% of their time in classrooms, choosing instead to use their school visits to collect information and data (on mid-day meals, school infrastructure, follow up on schemes like uniforms and scholarships) demanded by their bosses.

An important consequence of this deeply entrenched input culture is that it legitimizes a narrative of performance defined solely in terms of responsiveness to rules and orders received from above rather than meeting any specific service delivery goals. Such a system makes dialogue, deliberation and problem solving redundant. This was evident when we observed meetings that CRCCs had with their superiors. The mandatory monthly meetings were entirely transactional. Superior requested followed up on progress related to administrative orders and explained requirements of new orders. Amidst these transactions, there was no space for deliberation – views are neither sought nor debated. After all, the main job is to follow orders. And in this atmosphere the real purpose that CRCCs are hired for: supporting classroom pedagogy, are ignored. For instance, all CRCCs were expected to fill a quality monitoring format based on their observations of teaching-learning practices. But these were never 'requested' or debated and discussed, giving CRCCs the clear message that academic mentoring was the least relevant aspect of their job. But even more important, this influences the skills and capabilities that officers develop in their jobs. Is it realistic to expect officers schooled in a culture of hierarchy and order following to develop capacities needed for 'academic support'?

The effects of hierarchy and rules are even more perverse when it comes to the classroom. An input culture only understands what Pritchett describes as 'thin' logistical tasks – like constructing a classroom, enrolling students. These are tasks that can be easily monitored through 'thin' information (formal qualifications, # of classrooms constructed). Performance in these systems is defined in 'thin' terms. In this culture, even the most complex, implementation intensive tasks or what Pritchett calls 'thick' tasks like teaching are reduced to thin facts which are used to determine performance. This is precisely what has happened to our classrooms. Easy to measure metrics, 'syllabus completion' and 'pass percentages' have held our classrooms hostage.

We've unconvered the extent of this in an ongoing study with Vincy Davis and Taanya Kapoor on secondary schools in Delhi.[3] Across a yearlong interaction with school teachers, we found teachers repeatedly expressing their understanding of performance in these terms. 'I am well aware of the learning levels of my students… but I still have to complete the syllabus in time,' said one teacher. Another described why pass percentages matter. 'In a meeting with a senior education department official,' he related, 'every head of school was asked to stand up and answer questions related to their (school performance) in the examination.'

This concern with syllabus completion and pass percentages is entirely a consequence of the fact that these are the two metrics that senior administrators use to measure teacher performance. But administrators aren't the culprit. The real problem is the organizational culture in which they are schooled which makes 'pass percentages' and 'syllabus completion' the only rational way of holding the system accountable for learning. This is the norm in a post office state.

Education reforms, especially efforts focused on improving learning and changing classroom transactions, have to engage with this input-focused post office state. In the ultimate analysis, it is the interplay between reform ideas and the post office state that will determine whether a reform effort will succeed or, as the IAS officer reminded us 'rest in peace'!

Changing system behaviour, is not, as many reforms have tended to assume, a simple matter of changing the metrics of measurement, changing incentives through pay for performance, introducing new pedagogy or even disciplining teachers to ensure they show up, even though each of these are sensible reforms in their own right. For reforms to work, they have to directly address the mindsets and perceptions of frontline actors and how this may influence the way reforms are internalized and implemented.

In our work in Delhi, where the state government is engaged in a unique effort to shift classroom pedagogy and teacher orientation away from pass percentages and syllabus completion toward improving learning, teachers and administrators were flummoxed by reform expectations, at least in the first year that changes were introduced. The primary reform instrument was to reorganize the classroom according to learning levels so that teachers could teach students who were significantly behind curriculum expectations in a manner that allowed them to learn key concepts and eventually catch up.

Teachers were given the flexibility to move away from the syllabus and use alternative pedagogical techniques in these new classrooms. But teachers, whose experience was shaped by a system that required them to teach to the syllabus, found it very difficult to adapt to a flexible and somewhat more 'learning' focused system. In the end, our classroom observations found very little difference in teaching practices adopted in the different classrooms.

In fact, in an effort to facilitate 'differential teaching', toward the end of the year, the government reduced the syllabus to enable teachers to teach students at a slower pace. We found in the classrooms observed, that most teachers used this as an opportunity to conduct revision classes to ensure maximize pass percentages! And it wasn't just the teachers. The education administration too found it difficult to adjust to a system that doesn't measure known metrics and despite repeated assurances by reformers about changes in performance metrics, the everyday interaction between teachers and administrator stuck to the familiar – administrators focused on pass percentages and other administrative targets and teachers complied accordingly.

This is not to suggest that change is not possible. Our research has served to substantiate the puzzle that the officials I quoted in the introduction to this essay presented. Pilots and mission mode programmes succeed. In Bihar, efforts to change classroom process by Pratham worked when the system went into mission mode. Simple, easy to achieve foundational learning goals were clearly articulated and the entire system aligned itself to support teachers with teaching tools, mentoring and monitoring of outcomes. New teaching methods and alternative textbooks were welcomed and adopted.

In Delhi too, the very same teachers who found it difficult to grasp the concept of adapting the syllabus to teach according to student learning levels, were able to very effectively respond to a political call to organize a two month long reading camp with the objective of ensuring that 'non-readers' in their schools achieve basic reading capability. Here too, teachers responded to the clear goal, time bound target and willingly took students outside the classroom and using new pedagogical practices, teach them to read.

The problem, just as the IAS officials who motivated our research described, is that when the expectation shifted from mission mode to incorporating change into everyday practice, the system defaulted back to business as usual. Teachers and bureaucrats, in both settings, argued that new methods could not be used regularly in the classroom because they are at odds with the syllabus, and learning gaps are a consequence of problems outside rather than inside the classroom. In the end, the post office syndrome resurfaced as soon as the mission converted to a daily routine.

Our research also points to important levers of long-term change. The pilots we studied in Bihar and Delhi were successful because efforts were made to shift management practices and disrupt the post office syndrome, even if briefly. In Bihar for instance the key to the pilot's success was a shift in leadership and training style. In the pilot phase, the district leadership encouraged active dialogue and problem solving with CRCCs instead of the usual practice of expressing hierarchy through orders and demanding compliance. This empowered the CRCCs and pushed them to recast themselves as active, empowered agents of change: 'We had direct access to the DM. We raised the issues... He was listening to us. With the DM's backing, we (CRCCs) felt extremely empowered.'

This change in leadership style was accompanied with direct hand-holding and regular on the job training which helped the CRCCs build the skills needed to implement the pilots successfully. Scaling this pilot, successfully would have required embedding these management practices and problem solving techniques into the education bureaucracy's mode of engagement. But getting this right would also require decentralizing the education bureaucracy so that it permits local improvements and focuses on support rather than 'thin' compliance.

The issue of performance monitoring and accountability for learning outcomes is far more complex. What are the appropriate 'thick' metrics of measurement that can monitor performance and build accountability without reducing teaching to meeting testing standards? What is the appropriate administrative design – centralized versus local – that can ensure autonomy in the classroom and independence to the frontline along with accountability for performance? There are no clear answers to these very complex and system altering questions. What we need, therefore, is a robust debate, rigorous research and a willingness to experiment with new systems design and management principles. But this will only be possible when reformers acknowledge the post office syndrome and are willing to engage with the frontline rather than dismissing it as peopled by an unruly, corrupt force that needs disciplining.

Transforming an entrenched system may seem an impossible task. But to expect a system that is designed and accountable for delivering inputs to cohere around the goal of learning without systemic institutional reforms is a recipe for disaster. Elementary education in India is at an important moment of transition. After decades of focusing almost exclusively on expanding the education system through focused input provision, the system has slowly taken a turn toward shifting the goalpost to improved learning. This is an important and radical change. But to leverage this transition, reformers have to tackle head on the much harder challenge of reforming the education bureaucracy. Only then will good ideas and great pilots succeed and sustain.

* This article draws on research conducted by the author and colleagues at the Centre for Policy Research's Accountability Initiative. For a more detailed analysis of the ideas and arguments presented here, see Y. Aiyar, S. Bhattacharya, 'The Post Office Paradox: A Case Study of the Education Block Level Bureaucracy', *Economic and Political Weekly* 51(2). March 2016, and Y. Aiyar, A. Dongre and V. Davis. Education Reforms, Bureaucracy and the Puzzles of Implementation: A Case Study From Bihar. IGC Working Paper, 2015, available on: https://www.theigc. org/wp-content/uploads/2015/11/Aiyar-et-al-2015-Working-paper.pdf

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2. L. Pritchett, The Risks of Education Systems from Design Mismatch and Global Isomorphism. Faculty Research Working Paper Series (RWP14-017), Harvard Kennedy School, Harvard University, 2014; L. Pritchett, Creating Education Systems Coherent for Learning Outcomes: Making the Transition from Schooling to Learning. Research on Improving Systems of Education (RISE) Working Paper, preliminary draft, 2015.

3. This is an ongoing multi-year in-depth study. Findings reported here are preliminary and pertain to the first year of the study (the 2016-17 academic year).

Betrayal or benefit?

RUKMINI BANERJI

FOR some years now, India has achieved close to universal enrolment for children in the elementary school age group. The decades long push to universalize schooling and the pressure of the Right to Education Act since 2010 has led to almost all children in the 6 to 14 age group being enrolled in school. This paper focuses on children who have reached class 8, the end of the elementary stage of schooling and also the point at which compulsory schooling ends.

Based on available data on this age group and on a set of recent empirical studies, the paper aims to shine a spotlight on what these children have achieved in terms of schooling and learning and on their likely future prospects beyond this stage. As learning becomes a higher priority in Indian educational policy and classroom practice, empirical work outlining the challenges of children acquiring academic and other capabilities needs to be placed at the centre of this debate.

Moreover, as a growing number of India's young people complete elementary education, the focus of educational policy has also shifted to secondary education. Fulfilling the policy objectives of universal quality secondary education for all children, established by the Rashtriya Madhyamik Shiksha Abhiyan (RMSA) in 2009, requires a careful look at the capabilities of children completing the elementary stage and the pathways available to them to continue their schooling.

The paper relies on three main sources of data. First, it uses estimates from the Annual Status of Education Report (ASER), an annual household based survey that generates estimates of children's schooling and foundational reading and arithmetic status that are representative at the district, state and national level. Each year from 2006 to 2016, ASER has reported on a sample of more than half a million children in the age group 5-16 across rural India.[1]

Second, the paper uses findings from a three-year study carried out in Nalanda district (Bihar) and Satara district (Maharashtra). A sample of approximately 6000 children from classes 6 to 8, representative at the district level, was identified and tracked between 2012 and 2015 in order to take a deeper look at the upper primary stage in India.[2]

Third, a study aiming to understand the transition from elementary to secondary school (class 8 to class 9) was conducted in two blocks each of Hardoi district (Uttar Pradesh) and Sambalpur district (Odisha). A census of all children in these four blocks who were enrolled in class 8 in 2014-15 was conducted; they were then tracked for 18 months (2014-16) to understand the issues and challenges they faced in the transition to secondary school.[3]

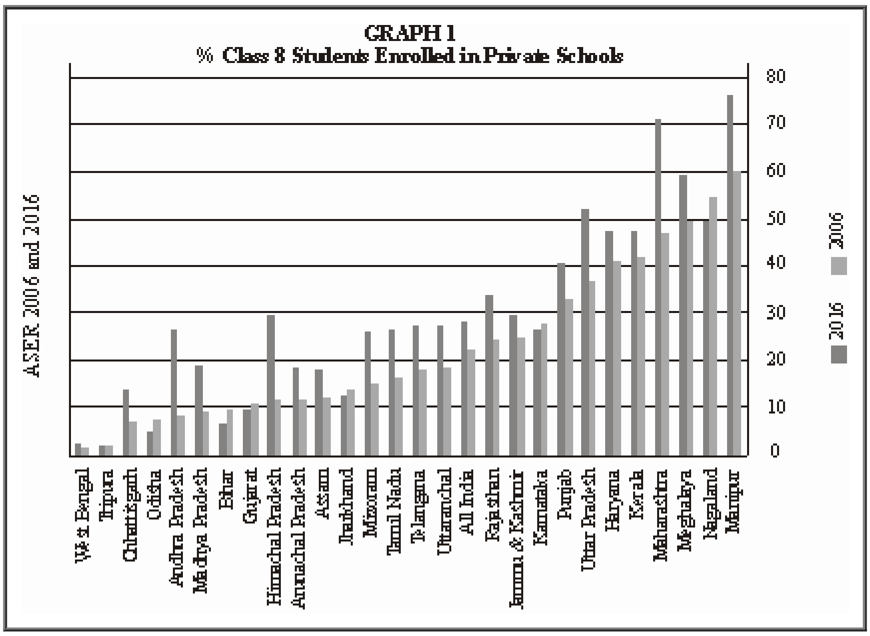
Let us set the stage for understanding class 8, the final year of elementary education in India. Who is in class 8, what kind of schools do they go to, do they attend, and where do they go afterwards? How have these patterns changed over time?

Size of student body: We begin with some back-of-the-envelope calculations. Based on census data, India has about 25 million children in each single age category. According to the District Information System for Education (DISE), ten years ago, in the 2006-7 school year about 12 million children were enrolled in class 8. By 2014-15, this number had risen to almost 22 million. Huge jumps are visible in educationally backward states like Bihar, where during this period, enrolment was a high priority for the government and incentives for elementary school completion were in place. For example, enrolment in rural government schools in Bihar in class 8 tripled from 5 lacs in 2006-7 to over 15 lacs by 2013-14.

Age-grade distribution: The RTE Act stipulates that education should be free and compulsory for children from age 6 to 14. In the legislation, elementary education is referred to as classes 1 to 8. In 2006, a nationally representative sample of rural children across India showed that only 30.5% of children in class 8 were 14 years old.[4] Another 35% were 13 years old. The remainder were either older than 14 (17.5%) or younger than 12 (16.8%). By 2016, the age distribution of children enrolled in class 8 had shifted to the left. Close to 20% were age 12 or younger, about 40% were 13 years old and only 11.8% were older than 14.

Decline in numbers of children not enrolled in school: Even in 2006, the overall enrolment figures for India showed that well over 93% of children in the age group 6 to 14 were enrolled in school.[5] According to ASER data, this number had risen to 97% by 2016. Within this category, if we look at the age group 11 to 14, significant shifts are visible: the percentage of boys (11-14) who were out of school dropped from 7.7% in 2006 to 4.1% in 2016 and for girls from 10.3% to 5.2%. The numbers for not-enrolled children age 15 and 16, ages not usually considered in elementary school statistics (even though they could still be studying in upper primary grades), dropped even more in this period: the proportion of boys fell from 20.2% in 2006 to 14.6% in 2016, and for girls from 22.6% to 16%.

Split between government and private school enrolment: According to ASER 2006, in rural India, 19.8% children were enrolled in private schools. By 2016, this proportion had increased to 30.6%. But a closer look at data over time indicates that the picture is much more varied than the national average suggests. For example, Graph 1 shows the percentage of class 8 students attending private school in 2016 as compared to 2006 for every state in the country. While there has been a large increase in private school enrolment in states like Uttar Pradesh, Himachal Pradesh and Maharashtra, there are also states where the change has been much smaller. In some states like Bihar and Odisha we see a decline in the proportion of class 8 children going to private schools, possibly because upper primary facilities also expanded during this period.



Continuation beyond class 8: Do young people continue their education after the free and compulsory schooling stage is over? The latest ASER survey from 2017 focused entirely on the age group 14 to 18. While not representative at either state or national level, data from 28 districts across India suggest that at least up to age 16, more than 85% of young people continue to be enrolled in school or college: over 60% in government institutions and about 25% in private schools or colleges. The remaining are currently not enrolled.[6]

The Hardoi-Sambalpur study provides a glimpse of the ways in which patterns of provisioning and enrolment beyond the elementary stage vary enormously in different contexts. For example,

* *Provisioning*: In the two Hardoi blocks, there were approximately 227 schools that offered class 8, but only 56 schools offering class 9 (most of which were private). In the Sambalpur blocks, the overall number of schools offering class 8 was much lower at 98 (mostly government upper primary schools), but a large proportion offered class 9 (33 schools, also mostly government).

* *Enrolment in class 8*: In the Sambalpur (Odisha) blocks, almost all children in class 8 were enrolled in government schools, whereas in the Hardoi (Uttar Pradesh) blocks, only 63% were enrolled in government schools.

* *Transition from class 8 to class 9*: The study found that in Hardoi (which has more schools overall and also a higher proportion of private schools), close to 40% children dropped out after class 8. The proportion of children dropping out was less than 8% in Sambalpur, which had far fewer schools and hardly any private schools.

These differences between Odisha and Uttar Pradesh are instructive. While Hardoi has more schools (hence potentially more choice and more access) than Sambalpur, the fact that most of Hardoi's secondary schools are private schools means that affordability is likely to be an issue. Despite having fewer schools (lower access and choice), Sambalpur children seem to move more easily between government upper primary schools and government secondary schools. These glimpses from empirical studies strongly suggest that as more and more young people complete eight years of schooling, we need a better understanding of how to prepare for the next step in terms of providing opportunities beyond the elementary school stage.

Attendance in school: While enrolment levels are high across the board and data is easily available, attendance is much more difficult to measure and hence attendance data is less easily available. What do we know about attendance, particularly for class 8? During the usual ASER survey every year, one government school with a primary section is visited in each sampled village. Hence, the sample of government schools with class 8 visited in the ASER survey may not be a representative sample of schools with class 8 in the district. Still, data from ASER 2016, covering more than 15,000 schools across rural India, showed that on an average day in the September to November period, about three quarters of the enrolled children were present.[7]

Since the Hardoi-Sambalpur study was a census of all schools with class 8 in the selected blocks, it provides an interesting glimpse of school attendance in these locations. Each school was visited twice during the baseline survey. In the Hardoi schools, a shockingly high figure of close to 60% of children enrolled in class 8 were absent on both days, in both government and private schools.[8] The situation in the Sambalpur blocks in Odisha was exactly the opposite. Although the total number of schools offering class 8 in Sambalpur was less than half of that in

Hardoi, well above 60% of enrolled students were in school on both days of the visit.

These examples suggest that there are important differences not only in provisioning and enrolment patterns across different contexts in India, but also that attendance patterns need to be studied closely, especially in upper primary grades like class 8 so that the links between attendance and learning can be better understood.

To summarize, the push for universalization of schooling has led to several pronounced trends that are visible at the end of the elementary schooling stage. First, more and more children in each cohort are staying in school longer; in recent cohorts, almost all children have continued till class 8. Second, over time the age range of the children reaching class 8 shows a shift towards younger children. This could have significant implications in terms of their preparedness for what lies ahead. If children enter school early (potentially before they are developmentally ready for class 1) and move automatically year on year through the school system, given the 'negative consequences of over-ambitious curriculum', younger children may be more likely to be 'left behind' in terms of learning.

Third, in terms of provisioning for class 8 and class 9, the patterns vary tremendously across India. The effect of the different patterns of provisioning on children's pathways forward needs much more empirical research. Finally, the same can be said of 'participation' in school. While enrolment data is easily available, much less is known about attendance or about any other measures of active participation in school. This is clearly another issue that needs more extensive study.

What about learning? Since 2005, the ASER surveys have provided annual estimates of basic reading and arithmetic for a nationally representative sample of children from close to 570 rural districts in India. Broadly, three clear trends are visible in the period 2005 to 2016.

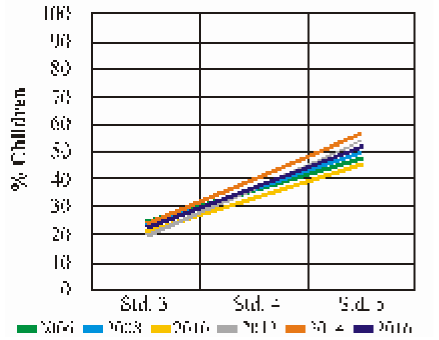
First, basic learning levels are low and have remained low over time (Table 1). This is the most often cited finding from ASER. More than ten years ago, about half of all children enrolled in class 5 in rural India were able to at least read class 2 level texts. This number was not very different in 2016; if anything, there are some signs of decline since 2010. Today about half of all children are completing primary school without foundational skills. Even after eight years of schooling about a quarter do not have basic reading skills: 27% children in class 8 were unable to read a class 2 level text in 2016.[9]

TABLE 1
% Children Who Can Read a Class 2 Level Text

Year	*Class 3*	*Class 5*	*Class 8*
2006	20.0	53.1	83.8
2008	22.2	56.2	84.8
2010	19.5	53.7	83.5
2012	21.4	46.8	76.4
2014	23.6	48.0	74.6
2016	25.1	47.8	73.0

Second, although children do acquire foundational skills as they continue in school and proceed to higher grades, their learning trajectories are relatively flat over time (Figure 1). If a goal of the school system is to ensure that most children reach grade level learning outcomes, then the learning curve for basic reading – a fundamental building block for future progress in school – needs to be much steeper during their primary school years.

FIGURE 1
% Children Who Can Read a Class 2 Level Test



Third, each successive cohort seems to do worse than the previous one. Ideally, to measure change in learning outcomes, the same children would be tracked over time. While ASER does not track the same children over time, it can be used to create artificial cohorts to see how successive cohorts are faring as they move through different grades. For example, a cohort that was in class 5 in 2007 would be in class 7 in 2009 and class 8 in 2010. Similarly, a cohort that is in class 5 in 2009 would be in class 8 in 2012, and so on.

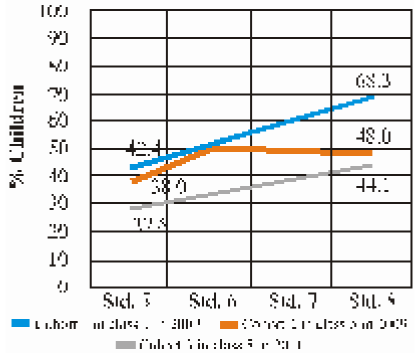
Table 2 presents learning outcomes in arithmetic of three such cohorts over time. Of the first cohort, 42.4% could do division (in class 5 in 2007) as compared to 38.0% of the cohort that was in class 5 in 2009. Of the children who entered class 5 in 2011, 27.6% could do so.[10]

Each column in Table 2 can be seen as a learning trajectory for a specific cohort of students. It is clear from the data that learning levels of each successive cohort in each grade are worse as compared to the previous cohort. For instance, by the time the 2007 class 5 cohort reached class 8 in 2010, 68.3% could do division. In contrast, 48.0% of the cohort that started class 5 in 2009, could do division by the time they reached class 8 in 2012; and 44.1% of the cohort that started class 5 in 2011 could do division when they reached class 8 in 2014. In other words, the learning trajectories of successive cohorts lie below those of previous cohorts (Figure 2). What this means is that each additional year of schooling is adding less and less for each successive cohort.

TABLE 2
% Children Who Can Do Division

	Cohort 1 in class 5 in 2007	*Cohort 2 in class 5 in 2009*	*Cohort 3 in class 5 in 2011*
Class 5	42.4	38.0	27.6
Class 6	50.0	50.1	33.1
Class 7	59.7	48.3	38.8
Class 8	68.3	48.0	44.1

FIGURE 3
Cohorts Over Time: Class 5 to Class 8
% Children Who Can Do Division



While the usual ASER survey is always a household level exercise with a one-on-one assessment of reading and arithmetic, the other studies mentioned in this paper use a combination of one-on-one assessments of foundational skills along with pen-paper tests. Depending on the study, up to 20% of students sampled even in class 8 could not read class 2 level text fluently. Basic levels in math were even more worrying.

The Nalanda-Satara study that tracked upper primary children found that if children did not already have foundational reading and arithmetic skills, they were unlikely to acquire them during the subsequent year in school. More than half remained at the same level even a year later. Not surprisingly, those who did not have foundational skills in upper primary grades were more likely to be female and from poorer and less educated families.

The Hardoi-Sambalpur research clearly showed that transition to secondary school was much lower among those who did not have basic foundations of numeracy and literacy in place.

The pen-paper assessments from both studies shared several common items. The tests were designed knowing that grade level learning outcomes were likely to be beyond the reach of many students in class 8.

Some results from the Hardoi-Sambalpur transition study are given in Table 3. They show, for example, that more students can do the easier operations (like subtraction) than the slightly more difficult operations (like division). Computations that need application and some critical thinking – like those using unitary method or calculating area and perimeter – pose more difficulty for students as compared to straightforward numerical problems. Calculating percentages, a task that is often needed in everyday life, was among the hardest.

TABLE 3
Hardoi-Sambalpur Study – Baseline and Endline Performance of Class 8 Children on Pen and Paper Test

		Hardoi N=1182		*Sambalpur N=852*	
Domain	*Examples of questions*	*Baseline (%)*	*Endline (%)*	*Baseline (%)*	*Endline (%)*
Subtraction: Numerical	426-251	59.3	59.4	58.0	57.4
		36.8	39.6	32.1	37.4

Subtraction: Word Problem	Anuradha read 18 pages of a book. If she read 10 pages more than Mukesh, how many pages did Mukesh read?				
Division: Numerical	950/20	31.7	36.7	29.0	27.2
Unitary Method	If 1 worker takes 2 days to make 8 boxes, then how many boxes can he make in 7 days?	37.2	40.9	43.5	40.4
Area	A rectangular field has dimensions 12m by 6m. What is the area of this field?	15.2	17.8	28.9	30.8
Perimeter	A rectangular field has dimensions 12m by 6m. What is the perimeter of this field?	5.0	6.1	23.6	18.7
Percentage	The total marks on a Math test are 120. If Sunil has scored 40% marks on this test, how many marks has he scored?	9.3	11.9	20.2	19.8

Further, children made little progress on being able to do these tasks in the year that elapsed between baseline and endline. A similar pattern of results is seen in the Nalanda-Satara study. These data – drawn from two separate studies in four states – suggest that despite the variation across study locations, even in class 8 children are well below the level of understanding and ability expected of them in primary grades.

In many ways the education discourse in India is beginning to acknowledge the learning crisis. For example, at the policy level, in 2017, the section on curriculum and completion of elementary education in the Right to Education Act (2010) was amended to include 'the preparation of class-wise, subject-wise learning outcomes for all elementary classes.' In the media as well as within the bureaucracy, there has been concern about the burden of the syllabus that children have to cover in school. At the same time, there is an increased focus on different aspects of secondary education that range from constraints in provisioning, to issues related to teacher shortage and quality, to governance and leadership, motivation, incentives and other matters.

But if adequate preparation of students is the key issue especially in the context of completion of the elementary stage, then any effort to reform or reimagine secondary education must acknowledge and deal with the weaknesses of policies and processes in the primary and upper primary years.

Several key challenges surface repeatedly from the research discussed in this paper. First, a substantial proportion of class 8 graduates do not have basic literacy or numeracy skills. The majority of class 8 students are still struggling with tasks that are expected of children in class 5. When students have not acquired capabilities expected of them in primary grades, it is difficult to ‘catch up’ in later years. Given that much of the teaching in Indian classrooms targets children who are at the top of the class; those who are far behind do not get any opportunity or support in school to catch up.

Evidence from these studies shows that the improvement in mean scores in language or math is minimal even after a year of being in school, regardless of whether the child is in upper primary or even when he or she has already transitioned to class 9. Those who are doing well academically move further and further ahead and the gaps between students grow as they spend longer in school. Being in this kind of ‘low learning trap’ means that although there is expenditure on schooling by families and by the government for each year spent in school, the ‘value added’ in terms of learning is minimal. The low productivity of this expenditure is not just a waste of financial resources but also has huge implications in terms of the mismatch between rising aspirations of parents and students and realities on the ground.

The way forward can be thought about in two ways – first, what needs to be done immediately; and second, a set of actions in the medium term. Within each, it is useful to think about what else is needed to help us understand the situation as well as what needs to be done to provide solutions to the problems that we identify.

The studies highlighted in this paper point out the need for further research on pathways through school. The macro trends of provisioning show that there is enormous variation across India in terms of where children go to study, how much they learn and what prospects they face. Today, well over 20 million children are completing the compulsory stage of schooling each year. We need to know more about the process that brought them there and the pathways ahead.

There is a lot of activity in the area of school assessment and student achievement in India today. There is a growing realization that the teaching-learning activities in school are not leading to grade level learning outcomes. It is worth noting that the National Achievement Survey of 2017 (which surveyed classes 3, 5 and 8) makes some interesting shifts from prior practice. The tasks are not curriculum based – instead they are competency based. Each test also tests competencies that are linked not only to that grade but also a few grade levels below. Finally, the report cards that have been published so far present data at district and state level.

But two further points need to be acknowledged in any assessment exercise in India. First, the fact that substantial proportions of children cannot read fluently means that pen-paper tests are not appropriate. Children’s low level of performance (tied to their lack of foundational skills) goes unnoticed. And second, the reporting of test results should be such that it can be easily translated into action by district education officials and teachers.[11]

An 'over-ambitious curriculum' and the usual practice of 'teaching to the top of the class', both contribute to the current situation. At a policy level, serious rethinking not just about the quantum of curriculum but also the content, sequence, pace and relevance needs to be debated and discussed widely. It is imperative that as a country we think through and define the capabilities that a child should have acquired by the time he or she completes the compulsory stage of education. The Right to Education must guarantee a meaningful 'completion certificate' in terms of learning achieved over eight years of schooling.

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Footnotes:

1. For a summary see ASER trends over time, 2006-2014. Available online at http://www.asercentre.org/Keywords/p/236.html

2. Henceforth Nalanda-Satara study. Policy brief available at http://img.asercentre.org/docs/Home/Homepage/middleschool sinindiapolicybriefjan2018.pdf

3. Henceforth Hardoi-Sambalpur study. Most research studies take a representative sample of children from a given geography. In this study however, a complete census of all children enrolled in std 8 was attempted in all four blocks. See http://img.asercentre.org/docs/Research%20and%20Assessments/Current/Education/Research%20Projects/Studyon AccessTransitionandLearningin Secondary EducationFull ReportJanuary2018.pdf for more information.

4. ASER data 2006 to 2016. See http://www.asercentre.org//p/289.html

5. This estimate matches almost exactly with estimates from IMRB surveys commissioned by MHRD which showed that in 2005, the proportion of out of school children (OOSC) in India was 6.9%. See IMRB, 'National Sample Survey of Estimation of Out-of-School Children in the Age 6-13 in India', Social and Rural Research Institute, IMRB, India Market Research Bureau and EdCil, Educational Consultants India Limited, Delhi, September 2014.

6. ASER 2017 data shows that 21% of seventeen-year olds and 31% of eighteen-year olds are not currently enrolled.

7. Any average figure masks vast disparities. For example, in the case of school attendance data from ASER 2016, states falling below the national average of 73.2% attendance include Telangana and Rajasthan (attendance levels between 73.2 and 70%), but also Tripura and Jharkhand (between 60 and 70%); Bihar, Uttar Pradesh, and Manipur (between 50 and 60%); and West Bengal (far behind with only 45.7% of children present in the school on a given day).

8. The attendance figure for Uttar Pradesh's government upper primary schools from ASER 2016 is among the lowest in the country at 55.8%.

9. The decline in learning levels that was initially highlighted by the ASER data is also is visible in other analyses. See for example, comparisons of the government's National Achievement Survey data for class 5 over time (eg. NCERT (2015), *What Students Know and Can do: A Summary Report of India's National Achievement Survey: Class 5 (Cycle 4)*, 2015); G. Kingdon, S. Sinha and V. Kaul, *Value for Money from Public Expenditure on Elementary Education in India*. South Asia Region, Education Global Practice, World Bank, New Delhi, 2016; analyses based on the longitudinal Young Lives study in Andhra Pradesh, among others.

10. The analysis includes children in both government and private schools. A similar analysis of only government schools shows that learning levels are lower as compared to private schools. Two points need to be kept in mind in doing such comparisons. First, it is well known that the demographic and background characteristics of private school children can be quite different from those of government school children – these need to be controlled for when comparing learning outcomes. And, second, even children in private schools are far from reaching grade level expectations.

11. The 2017 amendment to the RTE Act also stipulates the preparation of 'guidelines for putting into practice continuous and comprehensive evaluation, to achieve the defined learning outcomes.' The past record shows that CCE has not led to any significant change in practice. Hence education departments will have to work out how assessment can inform what teachers actually do in classrooms.

Board exams, rote learning and the learning crisis

SRIDHAR RAJAGOPALAN

THIS article starts by explaining why the author believes there is a learning crisis in India, how it probably originated and why it has persisted despite resources expended over many years. It goes on to discuss rote learning, seen as one of the root causes for the learning crisis, and board exams, which sustain rote learning. Finally, it talks about some proposed solutions.

Multiple studies have shown that there is a clear gap between what we expect children to know and understand, and what they are able to do. The reason we consider it a crisis in India (rather than just a learning gap) is that both in relative and absolute terms, the situation is extremely serious. India stood 73 out of 74 countries in the 2009-10 PISA test. Meanwhile, the ASER tests show repeatedly that about 50% of grade 5 children cannot read a grade 2 text. The results in arithmetic are equally poor.

We live in a world where basic education and critical thinking are becoming prerequisites for most jobs. If a significant percentage of Indians do not acquire basic skills, this represents a danger not just to progress, but even peace and stability in the region and beyond.

What caused the learning crisis and why haven't years of effort and resources addressed it? To understand this, we shall introduce the ideas of *technical and administrative paradigms*.[1] Every sector needs solutions in both the technical space and administrative (or governance) space. To take telecom as an example, the various core technologies (like 4G and 5G) relate to the technical paradigm. Practices like the process to allot spectrum correspond to the governance or administrative paradigm. A sector can flourish only if there is focus and progress on both these paradigms in that sector.

In public education, the focus seems to have been disproportionately on the administrative paradigm. The technical paradigm here refers to all aspects of curriculum, teaching techniques, research into learning and assessments, while the administrative paradigm covers areas like teacher accountability, teacher-student ratios, delivering textbooks on time, conducting examinations, and so on. There is an unstated assumption that the technical dimension is easy or trivial. This is seen both in comments like: 'What's there to teaching class 3 children – anyone should be able to do it…' and in administrative duties crowding out educational ones for many people working in the system.

We can now examine what sowed the seed for the current learning crisis. Our current education system was actually designed in the mid-1800s with a goal of producing effective workers and intermediaries to support the British Empire. While the world around has changed significantly and at an increasingly faster pace, education systems and curriculums tend to display a lot of inertia. The few changes that have been made (moving to the 10+2 system, semesterization, etc.) have tended to be based more on the administrative rather than the technical paradigm. The disconnect in the vision, in the language used and the lack of change over time have tended to make the enterprise of education/teaching more mechanical alongside a disproportionate focus on making it 'easier to test'.

This has led to what is called rote learning. We shall define and discuss it in depth later. Seen through the lens of the administrative paradigm, education is about children attending classes in which teachers are present, those children taking tests from time to time, and examinations at the end of the year. The very concept of 'learning' (as opposed to marks in a test) needs an appreciation of the technical paradigm.

Teacher training, in the administrative paradigm, is measured in terms of number of teachers trained and hours of training. This kind of training is inefficacious when the students are not from middle class backgrounds with educated parents and access to learning material. As the system expanded, a large number of teachers had to be recruited, but with training focused more on numbers, teacher quality suffered further. Today, when evidence of poor learning can no longer be ignored, it is teachers who are blamed for their poor skills. Of course, teachers have poor skills, but that's the symptom, not the cause of the problem.

How does teacher capacity building differ when seen from the lens of the technical paradigm? The focus would be on what training programmes are being offered, their impact if any, and correlating the actual learning gaps with those teacher training programmes. In many states, it has been observed that Block and Cluster Resource Coordinators do not play an academic support role (the original intention) but end up with administrative responsibilities – a classic example of the administrative crowding out the technical.

Let us now turn to the issue of rote learning and its role in sustaining the learning crisis. Reforming an education system is not easy, especially when the problems have accumulated over decades. However, there is one key lever that is available in tackling the learning crisis that we face today, and that is addressing rote learning.

What is rote learning? Many people think that memorizing or learning things by heart is rote learning. Rather, rote learning is learning something meaningless or which the learner does not understand, only or primarily because it is 'required' to be learnt for a test or because the teacher expects it. If one memorizes multiplication tables (and understands what '5 7's are 35' means), it is not rote learning and actually improves your arithmetic fluency. But if one learns a formula or a definition by heart without quite understanding it, simply because it will be asked in the test, that is rote learning. If a teacher asks students to name three things made of wood, and then expects them to name the same three objects mentioned in the textbook, that is rote learning. If students can recite the definition of a triangle, but cannot identify a narrow three-sided figure as a triangle or if they memorize an answer in English literature or History containing terms they don't understand, those are examples of rote learning.

People sometimes argue that though bad, rote learning is not as harmful as made out. We disagree. Once rote learning starts dominating a system, it kills meaning and purpose, not just for students but even for teachers. Learning can be a most exciting and passionate endeavour. Rote learning kills that curiosity and passion. What begins as rote learning in a chapter in Mathematics or Language becomes a system of accepting what a textbook or a teacher says, even if it does not make sense. It discourages thinking and encourages accepting things either because others accept it or because 'that is the way things are always done'. This expands into a rote system where marks, certificates and jobs are valued, and not real life skills. This is the perfect formula for dividing things between 'what happens in textbooks and the school' and 'what happens in real life', a distinction that is very clear in today's world (for example, 'speak in mother tongue at home but in English in school').

The opposite of rote learning is 'learning with understanding'. Sometimes people mistakenly believe that learning with understanding is merely a higher level of learning beyond rote learning. But students cannot attain rote learning first and then 'learning with understanding' because very different skills are needed and developed in each of them.

Many countries complain about rote learning in their educational system. However, its extent and intensity are extremely high in India. For example, consider a common question in primary school Mathematics – calculating the least common multiplier (or LCM) of two numbers like 12 and 20. In many school systems, children follow a procedure to reach the answer of 120. It is also likely that in many of these situations, children do not fully understand the significance of what they are doing. The difference we have found is that in India (and some other countries), teachers accept this and say, 'This is okay, we all learnt it that way only'. But in other countries, less tolerant of rote, teachers acknowledge that students often do not understand without implying that it is okay.

In 2006 and 2010, our company, jointly with Wipro, conducted studies which were published in *India Today*,[2] *The Hindu* and *Mint* publications. These studies showed that the problem of rote learning affects even our 'top' schools, and the performance of grade 4 students in these top schools was below international average. Further, the situation did not improve in the intervening years.

The cause and persistence of rote learning can be ascribed to one key attribute – our board exams, which influence all our school exams. On the positive side, this means that a lot can be achieved by reforming the pattern of questions in our board exams.

People are sometimes surprised to know that board exams can make learning more rote-based in the primary classes. We observed this phenomenon when we started a school, initially only up to class 5 and then added one higher class every year. In the initial years we found the teachers and classes focused on learning with understanding. But once teachers had to prepare their students to write a board exam, there was a marked shift which had a clear downstream effect. Teachers of class 9 would expect children to be prepared in a certain way in class 8 to be able to answer board questions well. These included things like 'not writing the answer in one's own words', 'using simple language', 'using terms and expressions from the textbook where possible', 'using examples given from the textbook', 'sticking to the information in the textbook even if it was wrong or outdated' – perfect recipes for rote learning. This was the exact opposite of what we would normally tell teachers and students, and this is how rote learning in the board exam (which is the 'goalpost'), affects the system as a whole.

There are serious problems at every stage of the board exam and all contribute to the learning crisis: the very validity of questions in the board exams – the NCF 2005 included position papers on many different topics. Here is an excerpt from the paper on examination reforms 2004-05:[3]

'The core of the exam system is the exam paper. This may seem almost a tautological assertion but, given the lack of attention paid by most boards to the quality of the actual exam paper, it is necessary to make it. The question papers remain seriously problematic.'

Question paper sets from the most recent (March 2004) 10th and 12th grade exams were collected for detailed study. Attention was focused on paper sets from five boards popularly perceived to be the best in the country. The exercise was an eye-opener:

* *What is the weight of the pituitary gland*? (non-essential, the gland should be studied for its crucial function, perhaps even for its structure, but hardly its weight!)

* *Who were the parents of Benito Mussolini*? (Irrelevant.)

* *How many members are there in the U.N.O.*? (Transient.)

* *Who was called Modern Messiah*? (The term was employed by the textbook writer to describe Karl Marx but has no wide currency outside the textbook.)

* *Describe the method of irrigation prevalent in India* (Takes as a given fact that there is only one such method in India, perhaps because the textbook has mentioned only one!)

* *Our highest import is from (Hong Kong, Italy, Kuwait)*? (No correct answer provided – at least for any year in the last quarter century.)

Unfortunately, those criticisms of 2005 are equally valid today.

The problem of poor and invalid questions does not plague the board exams alone. As part of an NCTE committee examining the reasons for only 1% of candidates passing the Central Teacher Eligibility Test (CTET), I had the opportunity to analyse questions from the November 2012 CTET test along with the performance data. Note that CTET is set by the CBSE (and is considered a much better test than the state TETs). Here are two questions from the paper.

a) One Pashmina shawl is as warm as ...... normal sweaters and .....hours are taken to weave a plain Pashmina shawl:

(i) 10; 200; (ii) 06; 250; (iii) 06; 200; (iv) 10; 250

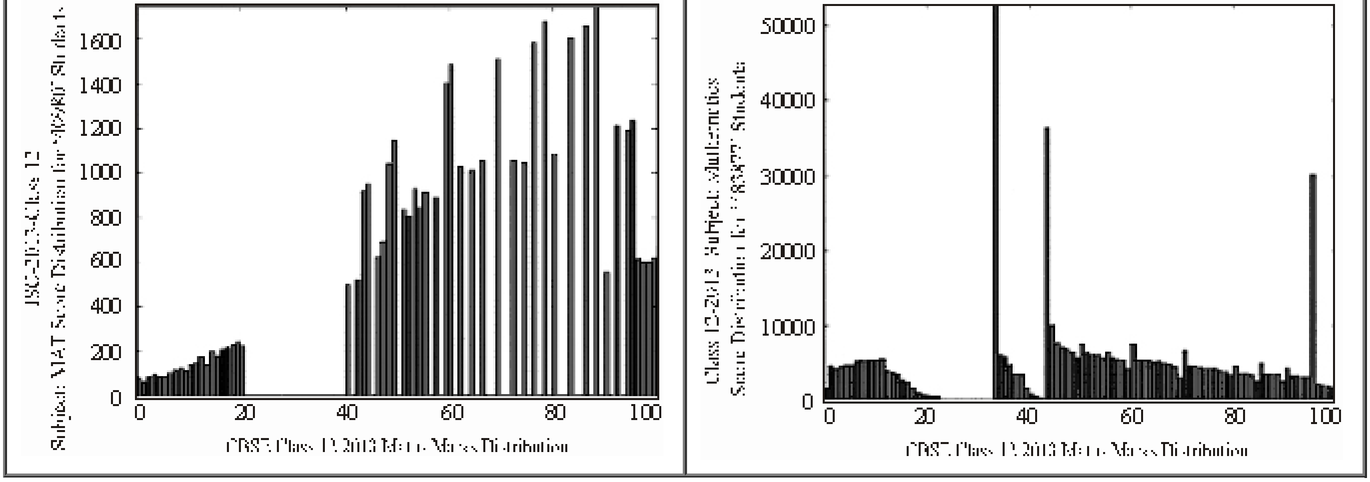
b) In order to instil a positive environment in a primary class, a teacher should:

(i) wish each child in the morning; (ii) not discriminate and set the same goal for every child; (iii) allow them to make groups on their own on the basis of sociometry during group activities; (iv) narrate stories with positive endings.

The basis of the data for the first question (apart from how comparative warmth is measured) is unclear. In the second question, the correct answer was marked as (1) and experts commented that most of the other options seem better ones. Let me add that I have not cherry-picked questions, a majority of questions had issues with validity or ambiguity or level of difficulty.

Somehow this issue of quality and appropriateness of questions in exams has never got much attention, nationally or internationally. Internationally, this is not seen as a big problem in the developed world and researchers there probably assume that questions would be fine. However, recently a working paper by Newman Burdett compared board exam papers from India, Pakistan, Uganda, Nigeria and a Canadian province, Alberta and found that the extent of rote questioning in India and Pakistan was even higher than the African countries: 'In India and Pakistan, higher-order skills were almost entirely lacking and the focus was very much on recall of very specific rote-learnt knowledge… In the two African countries, this rote-learning approach seemed less extreme.' [4]

The process to set and check the papers and keep them error-free: The process of making papers has remained a largely manual process where question makers submit papers in sealed covers and a paper is chosen at random for the actual test. These days, it is possible to use software to have questions solved and checked independently so that errors and even ambiguities can be all but eliminated.



The way board exams are scored. The graphs below from the work of Prashant Bhattarcharji[5] show how the marks of ISC (left) and CBSE (right) students in the Maths class 12 exam in 2013 are distributed. This is data for every student who took the tests.

There are two points to note – each of which represents a serious problem if not a scam. (*i*) These are not normal curves, which is very odd and would need explanation. Notably all spikes at marks like 36, 43 and around 96 in CBSE and 88, 95 and other scores in ISC are inexplicable and unfair. (*ii*) Both graphs show gaps, meaning no student scored those marks. This is not possible under normal conditions and suggests grace marks, also unfair to other students.

Why is this happening? It is said, 'Never attribute to malice that which is adequately explained by stupidity.' We see this more as a lack of capacity in the boards rather than a desire to manipulate. Yet, this is extremely unfair for a high stakes school leaving exam. It must be noted that nothing much changed based on these revelations in 2014; the author detected the same irregularities the next year.

There is also the issue of comparability of board marks. One can understand that marks of two different boards are not comparable. But for many years, even marks across CBSE zones were not comparable (students score higher in the Chennai region of CBSE compared to the Delhi region). These are apparently done to allow the students to compete with the local state boards but are unfair for students who seek admission in a different region.

Earlier, we have discussed in some depth why we feel there is a learning crisis, its original causes and why it persists despite so much expenditure of effort and resources. We have also discussed the key role of rote learning and many board exam related problems. In this section, we shall discuss some solutions.

The first principle that we should remember is that deep problems usually do not have quick fixes. All the solutions proposed will take at least three to five years to have an impact. These solutions focus on fundamentally addressing key issues like rote learning and low capacity in the system.

Focus on foundational learning: For any student, learning is like constructing a building – there is a foundation and then there are stages with each serving as the base for the next. The foundation for all learning is that *by the age of 8 or 9*, children must be able to read fluently, do arithmetic operations and develop basic critical thinking skills. The age deadline (corresponding to class 3 or 4) is critical. Either by law or through a campaign (like Swachch Bharat), this should be set as a common goal to be achieved by states and all types of schools in the next five years (with intermediate goals). A number of enablers in the form of resources for teachers, books and apps for students and information for parents and society are needed to make this happen. This will include research and assessments. But this will be the first (and most important) step in the battle to defuse the learning crisis.

Build systemic and institutional capacity: Capacity building is often used to refer to training of teachers and educational personnel. However, this 'people capacity building' has to be preceded by 'institutional capacity building'. This essentially means that key educational institutions – research institutions like NCERT, teacher training institutions like colleges of education and DIETs, assessment institutions like the boards and others like the NCTE – must be strong and should continuously strengthen their expertise in their core functional area. They should evoke respect from the academic as well as the practitioner community. The steps for this are appropriate leadership and hiring, research as a key role and providing autonomy to institutions. Do we, for instance, see NCERT in the same light as an IIT or IIM? Is that not possible to imagine and make happen?

Systemic capacity is the 'body of knowledge' that gets created and advances a sector. For example, how does one get children in very low income families to become effective readers? How does one create high quality computer based assessments in the Indian context? While some of this knowledge can and should draw upon what is already known internationally, there will be a lot of knowledge that will have to be created (and hopefully contributed to the international community). If Google can develop techniques of working with low bandwidth connections and offline maps in India (and then spread that to the rest of the world), can we not discover educational approaches that work in our conditions and use that to advance the field internationally?

Build a new Science of Learning: The Science of Learning is a new interdisciplinary science which draws upon school subjects like mathematics, science and language as well as pedagogy, psychology, cognitive sciences, neuroscience, AI and related fields to answer the question, 'How can human beings, especially children, learn better?' Just like medical science studies disease, symptoms, diagnostic techniques and treatments and builds on data to provide answers that doctors can use to treat patients, the Science of Learning uses data on learning, techniques for learning and teaching insights to provide insights to teachers on a regular basis. It allows teachers to use the art of teaching they practice and draw upon the Science of Learning whenever they need, to lead to improved student learning. Some Science of Learning researchers may study ways to improve reading skills in children, while others may research issues with Algebra learning in children, while still others may focus on effectiveness of AI based apps for students.

There is no doubt that we are in the midst of a learning crisis today. But we also have rich human resources that we can focus to not just solve this crisis for ourselves, but come up with techniques and solutions by strategically focusing on this area. Every country needs these solutions and there are, as yet, no ready answers. Can we convert this problem into an opportunity?

Footnotes:

1. Based on researches by Mariana Mazzucato including *https://www.ted.com/talks/mariana_mazzucato_government_ investor_ risk_taker_innovator*

2. https://www.ei-india.com/wp-content/uploads/2017/02/India-Today-printable-November-27-2006-Whats-Wrong-With-Our-Teaching.pdf

3. http://www.ncert.nic.in/html/pdf/school curriculum/position_papers/examination_reforms.pdf

4. https://www.riseprogramme.org/sites/www.riseprogramme.org/files/publications/RISE_WP-018_Burdett.pdf

5. http://www.thelearningpoint.net/home/ examination-results-2013/exposing-cbse- and-icse

The solutions lie with teachers

ANURAG BEHAR

IN early 2017 we published a research report about absence of teachers from public (government) schools.[1] The study found about 19% teachers were not in school. Of this 16.5% were out of school for perfectly legitimate reasons, such as, having been sent for training, attending meetings of the education department and having taken leave that was due to them. 2.5% of the teachers were absent from schools without any legitimate reason. It is these 2.5% who were playing truant. When teachers have to be out of school, such as the 16.5%, alternate arrangements should be made for students, and such occurrence should be minimized. But let us focus on the 2.5%.

The repeatedly brandished numbers in many quarters, including by many in education, of truancy of teachers, ranges between 25 and 50%. Though these numbers have no basis, they continue to be talked of as gospel truth. Our research study was one bit of systematic evidence against this falsehood. Teachers who read our report felt vindicated. Many on the ground workers in education and government officials felt relieved that a research study had found what they have been seeing all the while – that teacher truancy is nowhere near what it is alleged to be.

But many others, including some senior government officials, dismissed the findings. They had no basis to dismiss the findings, but somehow they ‘knew’ or ‘felt’ that it was wrong. In each such conversation we invited the people to gather evidence to support their ‘feelings’, perhaps conduct another research study. No one from this crowd has yet done that. A notable number of people working for ‘education reform’, i.e. to improve school education, also ignored the findings. This included researchers and civil society actors.

The findings of our study were not a new discovery. Most other studies on the same matter over the past ten odd years have found teacher truancy, or absence from school without legitimate cause, to be 3-4.5%.[2] Unlike our study, where we made categorical statements of the truancy (or absenteeism) number being vastly different from the popularly brandished numbers, most of these studies were inadvertently or advertently silent about the remarkable gap between reality and perception.

This silence has had a deeply damaging outcome. The headline numbers of these reports, which was about 20-28% for overall teacher absence (19% in our study) from schools, was construed to be the number for teacher truancy. This was then continually used as ‘evidence’ to support the notion that a very high proportion of teachers are habitually truant.

Given the history of this issue where matters that are amply clear are routinely obfuscated, let me state something explicitly. Teachers who are truant need to be disciplined and punished. They are personally responsible and accountable for such blatant transgression of basic professional norms.

Let us just call such teachers ‘bad’. Teachers who are absent from school for legitimate reasons are not the same as these truant or ‘bad’ teachers. If such legitimate absence has to be reduced, it needs interventions and decisions at the systemic level. For example, scheduling meetings after school hours, conducting training during school vacations, and recruiting and deploying teachers after factoring for leave available to teachers.

There is a staggering difference between the reality of public schools in India where 2.5-4.5% teachers are truant, versus the imaginary world of public schools that exists in the psyche of a vast number of people, where 25-50% teachers are truant. In this imagined world most teachers are 'bad', because that 25-50% number is surely the tip of the iceberg.

This matter of imaginary teacher truancy is emblematic of the deep narrative that these people have in their minds. The narrative is simply that the primary cause of all problems in Indian education is that the teachers are 'bad'; that a large majority are irresponsible, shirk work and have no professional commitment. Scapegoating the weakest in any system or society, and then embodying them as the cause of all ills is an often observed phenomenon. In India teachers are being scapegoated.

With this narrative of the 'bad' teacher being responsible for the ills of India's education, the solutions are also quite simple. The 'bad' teachers need to be fixed and told what to do. The specifics of the solutions vary – hiring only 'contract' teachers so that they can be easily fired; installing CCTV cameras in classrooms; having pre-packaged simplified content which teachers must transact with iron clad discipline; using technology to minimize the role of the teacher, are a few amongst many such solutions. The underlying principle of all these solutions and approaches is the same: teachers must be completely controlled and they cannot be trusted.

For the sake of brevity, let's call this the 'anti-teacher stance' (ATS). ATS helps simplify the complexity of education, albeit falsely, which then lends itself to simple solutions. It also helps absolve everyone else of the problems of education. Unfortunately, ATS is widespread in India, even if with varying intensity, including, by many officials who have a broad impact on education and by others working (supposedly) in school education for its improvement. ATS is unjust, educationally dysfunctional and strategically flawed.

It is unjust because the vast majority of teachers are not 'bad', though painted as so. It is educationally flawed because the teacher is at the heart of the social-human endeavour called education. Good education can happen only with engaged and high capacity teachers. It is strategically dysfunctional because it makes the teacher, who is actually the front-line of education, into the enemy. This is like going into a war while believing that your own frontline soldiers are the enemy.

All problems in Indian education are not because of ATS. However, a significant explanation for the lack of real improvement in educational outcomes is ATS. That is because ATS does not permit most interventions for improvement to energize and engage the teacher in the efforts for improvement. In fact, often such initiatives alienate and demotivate the teacher, because the very approach and design takes away the teachers' agency and attempts to closely control her.

At a deeper level, ATS filters out some of the most important issues in school education. This happens because once the teacher is identified as the focus of the problems, there seems no necessity to delve any deeper.

For India to truly improve its education, we need to move from an anti-teacher stance (ATS) to a pro-teacher stance (PTS) in all that we do in education. PTS will manifest in a careful and empathetic assessment of the reality of our teachers' challenges and circumstances. This would then inform all improvement interventions. As a result, one of the most important impacts of PTS will be that the existing eight million schoolteachers can be fully engaged and energized to lead improvement, instead of the current situation where they are at best on the sidelines and often alienated.

Once the system and its participants adopt PTS, then the priorities and actions will change, reflective of the root causes of the issues in Indian education rather than the convenient and false reduction of all problems to the teacher.

First, there will be a recognition that our public school teachers have extraordinarily challenging roles. While any teacher's role is complex and challenging, our public schoolteachers face very special and difficult challenges. Most of their students come from socio-economically disadvantaged backgrounds, many of them first generation school goers, and often from homes in acute poverty. Most teachers have to handle multiple grades at the same time and often multiple subjects. The schools do not have adequate learning resources and often lack adequate infrastructure.

Second, while the National Curricular Framework 2005 (NCF) is an educationally sound document, the curricula (across states) and its artefacts that the teacher encounters, is often deeply problematic. The syllabus, the textbooks and the examinations are poorly designed and do not reflect either the spirit or the principles of the NCF. The syllabus is content heavy, the textbooks are poorly written and the administrative culture pushes the teacher to 'cover' the textbooks. This fosters a culture of rote and the dysfunctional ritualization of educational processes, for example, of the recommendation for continuous comprehensive evaluation.

Third, our pre-service teacher education system is in shambles. It does not prepare our teachers adequately to perform their roles. The academic support available to teachers within the system is very weak and inadequate. The professional development opportunities provided to them are ineffective and desultory.

Fourth, the administrative culture treats the teachers poorly. It also loads them with demands which have little or nothing to do with their educational roles. The system/administration also keeps shifting priorities, moving from one project to another, most of which are superficial and driven by a shallow understanding of education. This continual shift of priorities distracts and confuses the teachers.

Given what our teachers face and deal with, let us turn to some possible actions. First, an adequate number of teachers need to be recruited and deployed. There must be subject-wise and grade-wise teachers. Schools with very small student numbers should be merged to enable this, though with caution and only where the access for students is not severely impacted.

Second, all schools should have basic infrastructure and adequate operating budgets. This includes toilets with running water, electricity and adequate number of rooms. The claim that 'input' conditions such as toilets have no effect on learning outcomes, and so we should not bother about them, can only be made by people who are disingenuous and disconnected at a human level. They should try working in offices that have no toilets, water and electricity.

Third, students from homes in poverty need a lot more support than teachers can provide. The minimal that needs to be done is to improve the meals provided in school. The budget allocations to the lunch must be increased and inflation linked. Breakfast (or a second meal) should be provided. Clusters of schools should be resourced with 'social workers' who work with communities and families to provide support, ensure attendance and minimize risks of children dropping out of school.

Fourth, curricula needs to be improved. And it must be done keeping in view the contextual reality of the teacher. This means that the textbooks and exams need to be improved. This has to be done across the states. Some states have made a reasonably good start on this.

Fifth, the pre-service teacher education system needs to be completely overhauled. This includes the regulatory approach, the curricula and the institutional approach. A large number of high quality 4-year integrated teacher education programmes need to be started in multidisciplinary institutions. All those teacher education colleges, which are educational institutions only in name, but actually shops selling degrees, must be shut down.

All these actions are equally important and must happen in parallel. Some will require increased public funding and some only political will. Some can be done with a little of both. Before detailing the last action in this admittedly non-comprehensive list, let me share some relevant experience of our work.

Our on-the-ground work is in some of the most disadvantaged districts of the country. In these districts we offer various kinds, called modes, of professional development opportunities for public schoolteachers. Let me use two examples of these modes.

On a couple of days every week, after school hours we organize a 60-90 minute discussion on topics of specific interest to teachers, for example, how to teach/explain the difference between perimeter and area or why corporal punishment has been outlawed. These discussions happen in what we call Teacher Learning Centres (TLCs). These are neat and clean rooms usually in the premises of a public school, with a small library of 1000-2000 books which are useful for the teachers, and a collection of teaching-learning material. We have such TLCs in about 200 small towns, where more than 150 teachers reside, across five states.

With over ten years of experience of doing this, we have observed that over 50% of teachers engage with such a mode. Of this, about 10-25% engage with great regularity. Our analysis suggests that the variation in these numbers is caused by the quality of the work we do, while the teacher populations and their characteristics are similar across all these places. All of this engagement is voluntary on part of the teacher.

A second mode is the 6-7 day residential workshops that we conduct during the summer and winter vacations. The relatively long time frame presents the possibility of going deeper into some matters. For example, how to bridge between the child's language and the medium of instruction, methods of history, use of local environment for science pedagogy, and so on. Again, participation in these workshops is voluntary for the teachers, meaning they *choose* to participate and there is no order from the government. In the five states where we work and offer these workshops, thousands of teachers participate, and yet we are not able to handle the number of teachers who want to register, which is many times more. Why do these teachers voluntarily invest so much of their time and energy in such activities? And this is a phenomenon that we observe across the country.

When asked that question the response is always the same – they want to become better teachers so that their students can learn better. This includes young teachers struggling with teaching basic language capacities and tenured ones wanting to do a better job. What I have called the 'pro-teacher stance' will harness and enhance this spirit, which we see across the country. The 'anti-teacher stance' diminishes it, and wastes this most important resource that already exists for improvement in education.

The last suggestion is that we must set up a vibrant system for continuing professional development (CPD) of teachers. If we want learning to improve, the capacity of our existing teachers must improve. Capacity development that improves effectiveness has a direct positive impact on motivation and engagement, sparking a virtuous spiral in the performance of their roles. A CPD system can have multiple institutional alternatives, including rejuvenated DIETs, BRCs and CRCs, and partnering with civil society institutions. However, any and all such systems to be vibrant and effective, must have the following characteristics and principles.

First, there must an explicit, clean break from the past, implying that the teachers and all stakeholders must understand that the torturous 'in-service training' that teachers have been subjected to is stopped and will not be repeated. Second, a high capacity team of 20-75 teacher educators will have to be carefully chosen and developed rigorously for each of the districts. Many of these people can be drawn from the teachers. These teacher educators will need a state level mechanism for their own continuing professional development.

Third, a CPD curricular framework must be developed to guide all CPD efforts. This will ensure that all CPD effort are meaningfully connected to specific objectives of professional development. And that there is connection and continuity across these efforts – which helps the teacher grow in a certain trajectory. Fourth, CPD opportunities must be offered in multiple modes – ranging from short and long workshops, to on-line reading and videos, to exposure visits, to in-school support and many others. All these must be informed by the curricular framework and must have continuity.

Fifth, these multiple modes must be offered in logistically convenient ways to the teachers. This is critical if the CPD efforts have to be continuous and not 'one-off'. Sixth, the sites where the CPD modes are offered must be welcoming places with clean toilets, decent places to sit and with reasonable arrangements for food. These things matter deeply over the long-term not merely because of the physical discomfort but also because the quality of these things manifest the respect that is given to teachers.

Seventh, while there must be many modes that are based on sessions led by expert teacher educators, there must be a sustained effort to foster and sustain local peer learning networks. The culture of these networks, which must be informal and supportive, will be the key to their success. Eighth, the overall tone and approach must be genuinely such that it sees the teachers as equal partners. The CPD system will become more and more robust as it engages teachers as true partners. Ninth, everything that is done within the CPD system must be designed to be interesting and relevant to the teacher. And the execution must be high quality. Tenth, the choice of what to learn, when to learn and in which mode must rest with the teacher. The teacher must have control over her own development trajectory, with her own learning objectives.

None of the above is easy to do or likely to yield quick results. But there is no other way to improve education but to work on the fundamentals. In fact, the reason that we are using phrases like 'crisis of learning' today is because over the past few decades, while the problems of our school education have been clear, we have refused to work on the fundamentals. Instead, we have tried an array of fundamentally flawed or superficial interventions, such as 'expansion of low cost private schools', 'technology for education', narrowly focused quick-win basic literacy programmes and so on.

All efforts to improve learning in schools will be mediated by teachers. So, improvement in learning in our schools will only track the capacity for improvement of our teachers and their engagement in such efforts. It is because of this that the teacher must be at the centre of all efforts, in a manner that energizes her completely.

Footnotes::

1. Teacher Absence Study, Azim Premji Foundation, 2017.

2. Ibid.

Learning to learn

MADHAV CHAVAN

NEWSPAPERS recently wrote about a crisis in engineering education.[1] It is reported that annually about 75,000 more engineering admission seats are going vacant because engineers are not getting jobs and that this is the result of poor quality of education. It should interest everyone that the nearly 72% of these colleges are private unaided that flourished over the last twenty-five years. But we do not hear many people talking about private sector being responsible for poor education. Anyway, the reason fewer people are going for engineering is because it does not end up in jobs and prosperity for a large number of graduates who are said to be unemployable. It is also true that with increasing automation, the industry needs either very highly skilled engineers or skilled shopfloor workers who need not be engineers. It would be interesting to see how many engineers are employed for their engineering degrees, if not knowledge.

Such exodus is not new. We saw it happen gradually out of municipal government schools in cities such as Mumbai. First, elite Marathi medium government-aided schools started losing students to English medium schools or started converting to English as medium of instruction. Next the municipal government schools that used to overflow with students once upon a time began emptying out starting at the southern tip of the city.

As we started measuring enrolment and reading/ math levels with the Annual Status of Education Report (ASER) survey, it became obvious that every year more and more children were moving to private schools. In 2016, as reported by DISE, the enrolment in private elementary schools is about 39%, which was just around 18% in 2007. It is not as though the private schools were there and children just switched. Just as the government was adding classrooms and appointing teachers, private operators sensed a new opportunity in the demand for education among the ordinary people as the economy grew rapidly. Private schools also started coming up in small towns and rural areas.

Today, more educated youth with high aspirations are finding it difficult to get jobs or get jobs that pay well. Many blame it on their lack of skills or lack of appropriate skills while others blame the economy. But what if masses of frustrated youth began to abandon the education system? ASER 2017, which surveyed young people between the ages of 14 and 18 found that although 85% were enrolled in schools or colleges, 40% worked for at least 15 days in the previous month. So, it is not clear what their enrolment means and how the institutions function. What if they were offered the alternative of not enrolling in institutions but instead in degree courses on-line, having access to notes and lectures (which do not change anyway), and directly appearing for examinations? It is not difficult to see that the colleges will empty out, although some may want to enrol so that they could hang out with friends. Or, perhaps there will be *dhabas* and cafes that will serve as places to hang out as they learn in groups? Then we will have a real crisis on our hands. But this is not possible. Or, is it?

It is commonly said that the economy is changing rapidly and the education system is not providing the kinds of skills to the children that are useful in the job market. But when did it ever? And how different are the foundational 21st century skills from what worked well in the 20th century? Barring the specific need to understand and handle the new and rapidly changing digital technology, the skill of learning to learn, adaptability to a situation, ability to communicate, solving problems, and team work were always marketable skills.

The difference is that the school system never focused on these outcomes of education and doing well in academic subjects was considered the prime aim of good education. In the lead was (and is) a fearsome system of examinations that gives people nightmares decades after they have been liberated from that tyranny.

The phrase ‘learning outcomes’ has begun to go around a lot these days. I think different people understand it differently but it is easy to see that it is not the same as inputs, which we were used to for decades. As early as in 1990, the Jomtien Declaration cautioned us:

‘Whether or not expanded educational opportunities will translate into meaningful development – for an individual or for society – depends ultimately on whether people actually learn as a result of those opportunities, i.e., whether they incorporate useful knowledge, reasoning ability, skills, and values. The focus of basic education must, therefore, be on actual learning acquisition and outcome, rather than exclusively upon enrolment, continued participation in organized programmes and completion of certification requirements. Active and participatory approaches are particularly valuable in assuring learning acquisition and allowing learners to reach their fullest potential. It is, therefore, necessary to define acceptable levels of learning acquisition for educational programmes and to improve and apply systems of assessing learning achievement.’[2]

But, ten years later in 2000, the Dakar Framework for Action seems to have no memory of this article. The 6th and last goal of the framework was about improving quality of education, which was to be measured by the survival rate till grade 5. Little surprise then that the MDGs, also written up in 2000, set the target for ‘Goal 2 – Achieve Universal Primary Education’ as ‘Ensure that, by 2015, children everywhere, boys and girls alike, will be able to complete a full course of primary schooling’ and the indicators were net enrolment ratios and survival rates till grade 5.

The ’90s seem to have set off a number of national and international student assessments in the western world. But there was no quantitative information on the issue of quality in the developing world which was the target of the Education for All campaigns. Pratham’s Annual Status of Education Report provided, almost inadvertently, the first quantitative data point about ‘learning outcomes’ indicated by ability to read and to solve simple numeracy problems at a very low floor level in 2005-06. Close on its heels but completely independently, in 2006 the Early Grade Reading Assessment or EGRA was born in the West. The important commonality about the two, apart from testing at very low levels, is that the reading assessment is conducted orally one on one with the child and it is not a pen and paper test unlike the others.

Over the last ten to fifteen years a lot of quantitative data on basic learning outcomes of reading and numeracy have become available around the developing world through the application of ASER and EGRA. In India, the work done by Education Initiatives has contributed to this information and more recently the NCERT has conducted massive pen and paper assessments with results available at the district level. Although none of these surveys and studies are comparable with one another, everyone agrees that they all point to poor status of basic academic learning outcomes.

The world seems to have suddenly become aware that learning out-comes are poor all around. The phrase 'schooling ain't learning' is now popular. Some decades ago it was said that years of schooling of its population is correlated with a nation's economic growth. Now there is data to support the thought that it is not the years of schooling but the assessment scores or learning outcomes that are correlated with economic progress.

The World Development Report of 2018 has focused entirely on 'learning' and said we are in a 'learning crisis' situation. Its three point action formula is:

First, assess learning to make it a serious goal. Information itself creates incentives for reform, but many countries lack the right metrics to measure learning.

Second, act on evidence to make schools work for learning. Great schools build strong teacher-learner relationships in classrooms. As brain science has advanced and educators have innovated, the knowledge of how students learn most effectively has greatly expanded. But the way many countries, communities and schools approach education often differs greatly from the most promising, evidence-based approaches.

Third, align actors to make the entire system work for learning. Innovation in classrooms will not have much impact if technical and political barriers at the system level prevent a focus on learning at the school level. This is the case in many countries stuck in low-learning traps; extricating them requires focused attention on the deeper causes.

I liked the World Development Report 2018 not only because it favours Pratham's kind of thinking, but because it is much broader and provides a good balanced picture of the situation and possible actions. It is available online for those wish to read it. All of us who are concerned about the status of education and learning outcomes are continually thinking of how to improve the system. It seems like this should be possible. But is it? Or is it too late?

This is the first time in the history of mankind that over 90% children in all countries are enrolled in formal schools with curricula, textbooks, teachers, classrooms, uniforms and examinations. This global rolling out of the mass education model, which was born in the late nineteenth century, has been happening over the last hundred, or at least seventy, years around the world. However, its efficiency and effectiveness was never questioned or measured seriously. It was assumed that if you open schools and provide all of the above, children will have the opportunity to get education, which was not incorrect. Clearly rise in literacy, availability of large numbers of educated people to perform many functions in growing populations and wealth were all outcomes of the expanding education system. There was reason for it to keep growing from the economic point of view and also from a social justice viewpoint.

The population of India was about 36 crore in 1951 and primary school enrolment was reported at 1.92 crore. Upper primary was 31 lakh. Today we have a population of about 130 crore and enrolment of 13.5 cr in primary and 6.5 cr in upper primary, making for an enrolment of around 96% children in this age group. In 2015-16, about 38% elementary schoolchildren went to private schools. In rural areas this proportion was around 30% and in urban around 57%.

This massive expansion of the school system in the far corners of our country may have had several objectives but as far as ordinary people were concerned it was all about 'getting education' to become a *padha likha insaan* and get a job, get out of poverty. The 'padha likha insaan' has some cultural socio-emotional attributes which should also count as outcomes of schooling but are not measured.

Ultimately, in the eyes of the *aam aadmi*, getting a good job is the main purpose of education. We have also been saying that if you get good education you will get a job. But is that true?

The table below shows how the number of students appearing for and passing std 10 and std 12 exams increased over a decade. Every year about 10 million students who clear their Higher Secondary School certificate examination enter undergraduate courses. Nearly 3.5 million go for BA, 1.5 million for BSc and another 1.3 million for BCom. These constitute about 60% of the students who come through the 'quality check' of the great education factory. Out of the remaining 4 million or so, 1.4 million seem to go to study various branches of technology and engineering. The remaining approximately 2.5 million new entrants go to some 50 odd courses including different branches of medicine.

Growing Number of Students and Improving Performance						
	Secondary Examinations			*Higher Secondary Examinations*		
Year	*Appeared*	*Passed*	*Pass %*	*Appeared*	*Passed*	*Pass %*
2005	1,34,86,740	86,09,855	64	78,28,610	53,86,899	72
2006	1,40,11,807	95,08,069	68	84,32,793	61,31,458	73
2007	1,47,12,429	1,03,44,687	70	88,68,155	65,66,178	74
2008	1,50,06,145	1,01,08,964	67	97,69,571	71,35,304	73
2009	1,70,89,658	1,18,19,689	69	1,05,51,871	80,44,636	76
2010	1,72,49,966	1,28,22,384	74	1,07,16,275	81,65,657	76
2011	1,80,24,161	1,34,89,768	75	1,15,67,858	87,11,254	75
2012	1,85,52,207	1,41,05,400	76	1,27,64,227	1,00,24,785	79
2013	2,03,15,906	1,57,97,305	78	1,40,59,950	1,10,34,079	78
2014	1,94,21,343	1,53,78,252	79	1,49,34,781	1,18,68,725	79
2015	1,91,16,429	1,50,89,278	79	1,43,23,497	1,13,41,644	79
Compiled from: http://mhrd.gov.in/statist						

In sum total, this is how the picture looks. Out of about 25 million children who enter std 1 every year, approximately 10 million or 40% will not go past std 10, one way or another. Of the 15 million who want to study more, half or 7.5 million will not graduate. Out of the 7.5 million who do graduate, how many are employable by the organized industry? It is said that 75% of fresh engineering graduates are not employable. So, we can easily take at least 50% as the not employable figure. That reduces the number of graduates employable in the organized sector down to about four million or less every year. This leads us to the next question – is the economy likely to generate four million new jobs every year that can absorb these graduates?

But this does not mean that the other 20-21 million remain unemployed. A large proportion may be underemployed. What is interesting is that government reported unemployment for those with school education is 3% while that for graduates is 10% and above graduation level is 18%.[3] This is very similar to the reported unemployment profile of the Chinese youth.[4] The numbers do not say if the employment is appropriately matched with the pre-employment training or not. In all probability it is not. I think, in most cases people with general abilities are employed who then learn the specifics on the job.

But what kind of jobs are these? Are these the kind of jobs for which you need to spend 10 to 15 years in schools and colleges? Have we been wasting everybody's time?

In the middle of all political sloganeering these days, have you noticed that nobody is promising jobs? The euphoria over 'demographic dividend' is not noticeable anymore either. It is increasingly becoming apparent that new job creation in the old way, whether in the manufacturing sector or the financial services sector, will be very difficult.

In fact, chances are jobs will be lost and with those lost jobs will vanish the main reasons for why children, especially older children, should attend schools and colleges in this education system. A new one will have to be invented or perhaps it will just come together as the old schools and colleges, government or private, irrespective of medium of instruction, will start emptying out.

I know what you are thinking: This cannot be. Sometimes I think that too.

A friend directed me to a remarkable article by Kai-Fu Lee, an accomplished artificial intelligence innovator, computer scientist, who is now a venture capitalist. I reproduce some portions from the article that helps me make a point.

'We're all going to face a very challenging next fifteen or twenty years, when half of the jobs are going to be replaced by machines. Humans have never seen this scale of massive job decimation…

'The industrial revolution took a lot longer. And the industrial revolution created jobs while it replaced jobs. When it took a few artisans months to create a car, an automobile, an assembly line allowed that to happen in a fraction of the time by dividing the work into little chunks. Some jobs disappeared. Many jobs were created. Car prices came down. And then the job employment rate went up.

'Artificial intelligence is different because when we make a (machine) loan officer that decides whether to give someone a loan or not based on purely quantitative information, that loan officer will be better than 99 per cent of all loan officers out there. They would be replaced outright because it is a simple, single domain optimization problem – feed everything we know about a person in, and out comes the likelihood of repayment versus default. That rate is a quantitative computation based on a huge amount of data that no human can possibly match. The people in those jobs will be out of jobs and will have to do something else – same with security, with paralegal, with accounting, and even with reporters and translators…

'…As we think about all the benefits from AI, there are a number of issues one needs to be concerned about. One issue, as I talked about, was the job losses and how to deal with that. Another issue is the haves and have nots. The people who are inventing these AI algorithms, building AI companies, they will become the haves. The people whose jobs are replaced will be the have nots. And the gap between them, whether it's in wealth or power, will be dramatic, and will be perhaps the largest that mankind has ever experienced…'[5]

If what Kai-Fu Lee predicts is broadly true, the current school system will be wholly inadequate for the education of the next generation. More than ever we will have to create social mechanisms for children's upbringing so that they can learn to face new challenges and to live in harmony with each other and with nature-environment. We cannot keep the school and the process of education isolated from the society. Teachers cannot be the only people responsible for children's education. In the truest sense, it will take the whole village to educate the child.

We are told that the ideal school system was meant to be for a holistic development of children. Most of the schools never came close to this ideal because doing well in the examinations was a priority. Now will be the time for the holistic school to come to life everywhere because being social, being creative, being caring and being human will be critical for everyone.

As jobs in manufacturing, financial services and so on are lost to the machines, in all probability activities in which human beings take care of others, where they entertain each other, where they engage in adventure and exploration, where they grow and create with their hands, will become important. Of course, we will need surgeons, engineers and researchers and they could be needed in large numbers. But, also visual arts, performing arts, athletics and sports along with foundational skills for learning and adapting will be important.

I have felt for some time that the non-linear character of the digital technology is reflecting in every sphere of our activity. It is reshaping our whole society once again. The industrial revolution once created a very organized society out of a disorganized agrarian life. Now we are changing again.

But, our education system refuses to budge from its assembly line-like, bureaucracy-influenced linear nature. What we learn, how we learn and when has to change. We do not need to discard and throw away everything we learn today. Learning to listen, read and comprehend will still be foundational. Learning to analyse solve problems will still be important. Above all, learning to learn, to be creative, to be human will have to be the most important universal outcomes of the next phase of education.

Footnotes:

1. https://economictimes.indiatimes.com/industry/services/education/india-is-in-the-middle-of-an-engineering-education-crisis/articleshow/63680625.cms

2. Article IV, Jomtien Declaration, 1990.

3. https://www.bloombergquint.com/union-budget-india/2017/01/23/budget-2017-few-jobs-for-young-india-youth-unemployment-is- twice-the-national-average

4. https://www.cnbc.com/2014/02/20/youth-unemployment-in-china-a-crisis-in-the-making.html

5. https://www.edge.org/conversation/kai_ fu_lee-we-are-here-to-create

Measuring learning outcomes in India

AMIT KAUSHIK

RECENT years have seen much debate about the crisis of learning. Study after study has shown that after spending billions of dollars and crores of rupees, we still seem to be grappling with lower and lower levels of learning in schools. Access has certainly gone up, and more children than ever before seem to be in school, but although there are many theories, it is not quite clear why the multiple interventions do not seem to positively affect learning. In a recent report, the World Bank points out that even after several years in school, millions of children cannot read, write or do basic math, referring to this learning crisis as one that is increasing social and economic gaps instead of helping to reduce them.[1]

A fundamental discussion, however, centres around the methods by which these learning levels are measured, and how reliable and valid data can be obtained. Clearly, accurate data is important to keep policy makers, school heads, teachers, and parents better informed, helping them to make more optimal choices about the way in which their children are educated.

Interestingly enough, policy documents in India prior to the 2000s almost never used the phrase 'learning outcomes'. The hoary Kothari Commission report, the National Policy on Education, 1968, and its successor in 1986 (including the Programme of Action, 1992), all dealt with various critical aspects of education, without referring to the outcomes of school education, except in passing or in the very broad sense of helping society to create democratic and aware citizens. Yet, almost every other relevant aspect of education – its objectives, structure and standards, teachers and teaching methodologies, enrolment, equity, curriculum, administration and supervision, even the physical location of schools – has been discussed in great detail.[2]

Even examination reform finds a place in these discussions – for instance, the National Policy on Education (1968) states in section 10 that 'a major goal of examination reforms should be to improve the reliability and validity of examinations and to make evaluation a continuous process aimed at helping the student to improve his level of achievement rather than at "certifying" the quality of his performance at a given moment of time.'[3] There is thus an indirect reference to learning outcomes here in the form of a 'level of achievement' but without further elucidation.

Similarly, the National Policy on Education (1986) refers to the need to lay down 'minimum levels of learning' – which in itself became the subject of controversy later when schools started to teach only to those levels – but does not go further to discuss their significance.[4] An entire section of this policy also deals with evaluation and examination reform, but restricts itself to viewing such devices as 'a measure of student development' and an 'instrument for improving teaching and learning'.[5]

Given the circumstances at the time, it is understandable that the focus was more on increasing access and ensuring equity. A poor country, recently freed from colonial rule, was seeking to build a more egalitarian society, and the emphasis, therefore, was on ensuring that children and young people received adequate and equitable opportunities for education, within limited resources. Even in the mid-1980s and early 1990s, resources remained a significant constraint within a closed economy even as the population had increased substantially, making it unsurprising that the objectives of policy remained consistently focused on access and equity.

Underlying this focus was an implicit assumption that the provision of adequate inputs would automatically lead to desired outcomes, even though these were not identified or explicitly stated. From the point of view of the bureaucracy this was certainly a preferred approach, since inputs are easily measured and accounted for, making it relatively simple to maintain the necessary official records. Clearly, the number of classrooms or toilets built or teachers appointed lend themselves to easy enumeration, as opposed to measuring an intangible like learning for which one would need different kinds of proxy indicators for latent abilities.

This approach was carried forward to the Right of Children to Free and Compulsory Education Act, 2009 (hereafter the RTE Act) which, at least initially, made no reference to learning outcomes.[6] It was only as recently as 2017 that the subsidiary rules were amended to provide for the preparation of class-wise, subject-wise learning outcomes,[7] in belated acknowledgement of the need to consider the end result of school processes.

In fairness though, and as someone who was peripherally a part of that process, I am aware that one of the main reasons why there was no particular emphasis on learning outcomes in the RTE Act was the introduction of the concept of Continuous and Comprehensive Evaluation (CCE) to accompany the no-detention provision. Even though it did not discuss learning outcomes specifically, the CABE subcommittee which drafted the 'essential provisions' of the draft legislation in 2005 was of the view that if CCE were done right, there would, at the end of each school year, be a detailed portfolio of each child, providing an exhaustive assessment of her achievements and abilities. It is another matter that this form of evaluation never materialized in practice, and that even if successful, would have provided information about individual students but not the system.

The change of public discourse to emphasize the importance of learning outcomes can reasonably be traced to the inception of the Annual Status of Education Reports (ASER) brought out by the NGO Pratham from 2005-06 onwards. Although the initial reports dealt primarily with enrolment, subsequent ones began to look at learning levels of enrolled children. As noted elsewhere by this author, despite the many methodological questions raised by some, this led to a shift in attitude, where inputs were assumed but outcomes became the focus of discussion.[8] Policy documents began to reflect this change, with the 12th five year plan noting that 'there is a need for a clear shift in strategy from a focus on inputs and increasing access and enrolment to teaching-learning process and its improvement in order to ensure adequate appropriate learning outcomes.'[9]

The spotlight on outcomes brings with it the need for accurate and reliable measurement, and the development of related capacity. Sporadic attempts at determining student achievements have been made by various researchers from 1970 onwards, but these rarely went beyond limited parameters. The introduction of DPEP witnessed the beginning of baseline, midline and endline surveys, but it was only as recently as 2001-2002 that NCERT institutionalized the current National Achievement Surveys (NAS) for grades 3, 5 and 8, and later, 10. Initial surveys were based on Classical Test Theory (CTT), so comparisons and analysis from year to year were not possible; lately, the surveys have moved to an Item Response Theory (IRT) base which allows monitoring trends over time and comparisons across the country and internationally.

For most countries, quality in school education goes hand-in-hand with the need to ensure equity. Yet, unlike ASER which is household-based, NAS does not include out-of-school children, and cannot, therefore, help address equity concerns. Further, except for the last cycle for grade 10, NAS does not include the performance of children from private schools and so the picture we receive remains incomplete, given that the latter account for nearly 40 per cent of overall school enrolments. Dealing with the learning crisis will need a more complete picture of learning than we currently have available to us.

It is important to emphasize here that the collection of learning data through large-scale external assessments should not be seen as replacing, or in opposition to, ongoing internal assessments that teachers undertake as part of the teaching-learning process. Instead, as Masters argues, the former serves as an important indicator of the health of the system, providing valuable information to policy makers to 'establish where learners are in their learning at the time of assessment's so that corrective action may be taken in time at a policy level.'[10]

Viewed in this light, even if done right, CCE as noted above would still remain an assessment of individual classroom performance and not be able to provide information on the well-being of the school system. Both types of assessments are clearly very different in their objectives and scope and should thus be seen as complementary, yielding valuable data that supports improvements in student learning.

The Sustainable Development Goals (SDGs) adopted by the world community in 2016 have a specific goal devoted to quality education. Departing from earlier practice, the SDGs speak of 'levels of proficiency', which clearly implies a need to measure such proficiency in a manner that is consistent and comparable within and across countries. The emphasis of the SDGs on quality of outcomes, rather than just inputs, immediately highlights the value of educational assessment and the critical role it can play in determining the extent to which a number of SDGs have been met. As a responsible member of the international community, India too will need to demonstrate her commitment to these agreed goals and to put in place mechanisms to gather the necessary evidence of progress.

However, one of the most significant difficulties in obtaining and using student learning data to plan for policy development and implementation, not just in India but also in the larger South Asian region, remains the weak institutional capacity available for this purpose. Technical capacity is nascent and fragile, and often insufficient for the magnitude of the task at hand. In our work with countries in the region, we have observed a number of challenges, some of which can become serious hurdles to accurate measurement of learning achievement.

Assessment of learning is a technical and highly specialized field. Yet, policy related to assessment continues to be driven by generalists with no particular training or expertise in the area. Unlike Nepal for example, which recruits officials with education-related qualifications to an education service and then continues to provide them with appropriate capacity building opportunities through their career, the erstwhile Indian Education Service was discontinued some time ago. A proposal to revive it was made by the now-defunct Subramanian Committee on the new education policy but those recommendations seem to have been placed in cold storage.[11] As a result, the Ministry of HRD at the Centre and education departments in the states continue to be manned by well intentioned officers from multiple civil services (but primarily the IAS), who usually lack the technical knowledge needed.

The lack of technically trained decision makers leads to several challenges. First, there is often little appreciation of the time and effort required in undertaking large-scale learning assessments, which in turn results in poor planning and implementation. A tick-box approach to assessment means that the emphasis is on 'doing' rather than on 'doing well'; significant resources are utilized on collecting data that is inaccurate or invalid because of incomplete planning or compromised quality processes. Often assessments may be carried out at more frequent intervals than needed, instead of taking enough time to get them right qualitatively, with the attendant cost implications. As an example, consider the proposal made by some officials of the Ministry of HRD a couple of years ago – a national learning survey collecting data *every* year about *every* subject, for *every* student, in *every* class from grade 1 through 8! Clearly, such proposals stem from a lack of technical understanding or appreciation of the purpose of large-scale diagnostic assessments.

The paucity of trained human resources is also a constraint in other ways. Within education departments, there are usually inadequate permanent resources allocated to assessment. What is often not realized is that conducting large-scale student assessments requires subject experts, assessment experts, statistics and psychometric experts, sampling experts, survey experts, language experts, graphic artists and designers, field administrators, and communication specialists.[12] All of these are rarely available at the same time to most education departments. Additionally, as is the case with other staff working in government departments, each one is tasked with a multitude of roles that frequently go beyond their primary responsibility for assessment.

Periodic transfers between government departments can also contribute to the lack of trained resources. Developing skills in assessment takes time but stability in tenures remains a challenge in the region at all levels. The head of the Education Review Office, Nepal, is on record in a published paper that frequent redeployment of staff members was one of the most difficult challenges faced in their national assessment.[13] Several states in India have created assessment cells to try to institutionalize processes and knowledge but are hampered by the fact that tenures of staff at such centres remain irregular.

A related issue is that universities in the region rarely consider including specific modules or courses on assessment methods as part of their pre-service teacher training programmes, other than perhaps topics related to in-class assessments. Specialized training in areas important to assessment, such as psychometrics, is almost non-existent, as a result of which the pool of expertise available to the public and private sectors is highly restricted. Some capacity has been built over the years within NCERT, but that is clearly not enough to support the vast scope of such assessments across the country.

The allocation of adequate financial resources to assessment remains a challenge, which can lead to compromises in processes or quality. In a recent assessment, due to budget constraints, the state decided to save on the cost of printing Optical Mark Recognition (OMR) sheets for students, preferring to capture their responses through manual data entry instead. As a result, even after several rounds of data re-verification and cleaning (which came with unnecessary additional expense), the state was only able to report the performance of about half of its assessed districts. In Afghanistan, learning data collected under the Monitoring Trends in Educational Growth (MTEG) project lay idle for nearly two years before being analysed due to resource constraints. Again, there is sometimes the perception that learning assessments can be carried out inexpensively, causing governments to underestimate the financial resources actually required to deliver them.

While the importance of measuring learning is widely understood, there are certain potential pitfalls that can reduce the utility of related assessments. First, there can be a real risk of moving from no measurement at all, to measuring everything. As Muller reminds us, it is important to remember that measurement is not an alternative to judgement. 'Measurement demands judgement: judgement about whether to measure, what to measure, how to evaluate the significance of what's been measured, whether rewards and penalties will be attached to the results, and to whom to make the measurements available.'[14] The massive increase in the scope and coverage of NAS in India in the 2017 cycle allows detailed reporting of learning outcomes – yet reporting is not the end objective here. The goal must be for states to *use* the data that NAS provides to help improve learning in the classroom.

Second, such external assessments are useful only if they are perceived as low stakes assessments by the children, the teachers, and the schools. In recent years, international agencies and governments are increasingly moving towards funding regimes that are linked to certain deliverables, including in some cases, improvement in learning outcomes. Similarly, there has recently been some discussion that under the newly restructured *Samagra Shiksha Abhiyan*, which has subsumed the erstwhile SSA and RMSA, some part of central funding to states may be linked to the School Education Quality Index (SEQI) which ranks states on a variety of parameters, including performance on learning outcomes.

Linking learning outcomes to incentives carries with it the risk that an external assessment (say, the NAS) would no longer be perceived as low stakes. Instead, the stakes would be raised for the national or state governments involved since part of their education budgets would be tied to the results, which in turn could lead to pressure on schools and teachers to show 'better' performance. Education history is replete with examples of what happens in such circumstances – the Chicago Public School cheating scandal in which teachers helped students on standardized tests was one such outcome of school incentives being linked to student performance.[15]

Any learning assessment is valuable only in terms of how it supports improved learning in the classroom. It is here that the link between the data emerging from a learning assessment and its implications for policy and practice needs to be strong. In 2009, two states from India, Tamil Nadu and Himachal Pradesh, considered to be among the most progressive educationally, participated in the Programme for International Student Assessment (PISA). An international assessment that includes more than 70 countries, PISA is conducted every three years by OECD to evaluate education systems worldwide by assessing the skills and knowledge of 15-year olds from participating countries in science, reading, and math. The two states ranked 72 and 73 respectively, in a field of 74.

One of the major reasons for this was the fact that PISA tests for application of knowledge, as opposed to traditional examination systems in South Asia that primarily test rote memorization. It is a well known fact that while memorization is useful for dealing with relatively simple questions, more complex concepts and questions require an ability to apply the memorized knowledge. The PISA results could have been used as an opportunity for India to review her assessment systems and move towards a less rote-based regime. Instead, we chose to withdraw from future PISA attempts.

The results of such large-scale assessments need to filter back into teacher professional development, changes in curriculum and teaching methodologies, and the entire approach to assessing learning. Strengthening this link requires investment in human resources, technical expertise, and institutionalizing capacities. India and other countries in the region have made a beginning in this direction, but a great deal more remains to be done; ensuring that our children learn well and are prepared for the future should demand that we make that effort.

Footnotes:

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Are India's schoolteachers prepared to teach?

TARA BÉTEILLE

TEACHERS are the most important school-based resource for improving student learning. But are schoolteachers in India equipped with the skills they need to be effective inside the classroom? Several indicators suggest otherwise. Low student learning levels are perhaps the most salient indicator, but there are other indicators that imply teachers are not adequately prepared to handle the complex task of teaching and classroom management.

First, there is growing evidence that many teachers lack the content knowledge needed to be effective. In a study of Bihar teachers, average subject knowledge of teachers was found to be weak, with the mean score in language at 45% and math at 58%.[1] Even when teachers displayed content knowledge, many could not explain concepts. For example, almost 80% of teachers had the correct answer to a long division problem (3 digit by 1 digit), but among them only 11% could do all the steps right.

Second, studies on teacher time-on-task suggest that Indian teachers spend most of their teaching time doing exactly those tasks associated with poor teaching: reading aloud, writing on the blackboard and barely engaging with their students.[2] Finally, nearly 27% of students avail of private tuitions at the primary level and more at the secondary level.[3] This is true for government school as well as private school students. Why students take private tuitions is a complex issue, driven by multiple factors – but it is unlikely their learning needs are being met in school if they need to go elsewhere to learn.

When teachers come poorly prepared to teach and manage classrooms, it is important to look at the pre-service education landscape that prepares and allows teacher-candidates to potentially seek a career in teaching. A good pre-service education programme is the first step toward equipping teacher-candidates with the content, pedagogical and managerial skills they might need for becoming effective teachers. That is how it is with other professions. For instance, one cannot be an effective doctor without having good pre-service medical education. Ironically, however, the quantitative evidence on the causal impact of pre-service education on a teacher's ability to improve student learning is inconclusive. Most studies find relatively little impact of a teacher's pre-service education on student learning; furthermore, there is limited consensus on which factors in pre-service education affect student learning more.[4]

Uncertainty regarding the effects of teacher pre-service education comes largely from three methodological challenges.[5] First, it is difficult to isolate teacher productivity, especially since a student's own ability, peer influences, and other characteristics of schools also affect measured outcomes. The problem is exacerbated by the biases resulting from the fact that assignment of students and teachers to classrooms is usually not random.

Second, there is an inherent selection problem in evaluating the effects of education and training on teacher productivity. Unobserved characteristics, such as motivation or intelligence of teacher-candidates, may affect the amount and types of education and training they choose to obtain as well as their subsequent performance in the classroom.

Third, it is difficult to obtain data that provide adequate detail about the various types of training teachers receive and even more difficult to link the training of teachers to the achievement of the students they teach.

Nevertheless, if we look at countries where teaching is a sought after career and student learning is high, a few things stand out about their pre-service education programmes.[6] First, entry into pre-service is highly selective. For instance, becoming a primary schoolteacher is highly competitive in Finland. Selection for primary schoolteacher education happens in two steps: first, candidates are selected based on scores in matriculation exam and out-of-school accomplishment records. Second, candidates take a written exam in pedagogy, their social and communication skills are observed in a clinical setting similar to school situations, and top candidates are interviewed and asked to explain their motivation to become teachers.

About one in every 10 applicants in Finland is accepted in teacher education programmes to become a primary schoolteacher. In Singapore, the government recruits the top one-third of high school graduates to enter teacher education programmes and does not require an entrance exam. In Korea, entrants into teacher education programmes are among the top 10% of high school graduates.

Second, once in pre-service education, candidates follow a curriculum that is linked to what happens in schools, and contain an extensive practical teaching component. In Finland, Korea and Shanghai, a practicum is required for primary and secondary levels comprising at least a six-month classroom teaching component. This allows teacher candidates to learn to apply pedagogical skills and gain skills in classroom management. It acquires special significance as teachers increasingly have to teach a very heterogenous group of students who are at different learning levels.

Third, pre-service education programmes in high performing education systems are closely linked with universities. This allows the curriculum to be informed by the latest research in learning and other fields, while also providing pre-service education a status similar to other undergraduate programmes. Finally, pre-service education programmes in all these countries are tightly regulated by the government. At the primary level, they are generally provided by the government as well.

In India, in contrast, the situation is very different, despite the landmark Justice J.S. Verma Commission report and recommendations presented in 2012. The report provides a clear road map for improving the teacher education system in India. But six years later, pre-service education programmes do not appear to be helping teacher-candidates become effective teachers any more than before.

The poor quality of teachers coming out of teacher training institutes is evident in the low pass rates in teacher eligibility tests. Less than 20% of teachers have passed the test in states where tests have been conducted. This suggests that the majority of those who take the teacher eligibility tests do not qualify to be recruited as teachers despite having the required academic degree and professional training in teacher education.[7]

When comparing India's teacher pre-service education landscape to well performing education systems, none of the pre-conditions are in place in India. First, pre-service education programmes are not selective. There are no entrance tests nor systematic processes for entry into such programmes. To put it bluntly, just about anyone can apply to become a teacher. Second, the curriculum of pre-service education programmes is divorced from the goals of school education as well as the realities of classroom practice.

At both the elementary and secondary level, the curriculum is fragmented and outdated, and does not address subject knowledge adequately. New developments in specific subjects are not incorporated. The focus is on general methods of teaching such as lecture, classroom discussion, question and answer, and memorization. Student teachers do not learn pedagogy skills.[8] Practice teaching in classrooms, for instance, lasts no more than five to six weeks and provides only piecemeal experiences of a fully functioning teacher.

A further concern in India, given the low selectivity of pre-service education programmes, is that teacher-candidates come from the same low quality schools where they will be heading to teach. There is little in the design of pre-service programmes to help remediate the academic deficiencies teacher-candidates bring with them, many of whom come from poor school systems, as they train to become teachers.

Assessments in teacher education programmes are of limited value in screening good candidates from weak ones. The current method of evaluating students enrolled in pre-service teacher education in India, for instance, does not sufficiently assess conceptual and pedagogic skills of teaching.[9] In addition, only cognitive skills are assessed. There is no assessment of student teachers' social and emotional and communication skills to engage with children.

Third, pre-service education colleges in India are typically stand alone, not benefitting from the latest research on teaching or links with other departments. At the primary and secondary levels, three types of teacher training institutes exist: government, privately aided and privately unaided. These include: at the primary level, the District Institutes of Education and Training in each state, which are fully financed by the central government, and at the secondary level, the Regional Institutes of Education that are also central government funded.

In rare cases, universities have departments of education. But for the vast majority, teacher education is located outside the realm of higher education, with most institutes located outside of university campuses.[10] This is often also true of those institutes that are affiliated with universities such as colleges of teacher education. The programmes are isolated, lack well defined professional standards, have low visibility, and do not benefit from new knowledge generated in universities.

A fallout of this is that the institutional capacity to prepare teacher educators is insufficient across the country, with too few programmes offering M.Ed. relative to the needs of specific states. The programmes are general and not able to address specific needs to develop subject experts. There is no policy for professional development for teacher educators.[11]

Finally, weak governance and ineffective regulations characterize the pre-service education landscape in India. There has been a significant increase in the number of teacher education institutes in the past decade. By 2011, the number of teacher education institutes increased to 14704 from about 1800 in 2008.[12] More recent official numbers are not publicly available. Most of this expansion has taken place in the private self-financing sector.[13] The tremendous growth in the numbers of teacher education institutes speaks to the unregulated nature of entry of these institutes: there are considerably more teacher-candidates being prepared than jobs available.

A key problem in managing quality is the relative lack of capacity to enforce norms and standards. This has led to wide variation in quality and a clutter of institutions and programmes in teacher education. As an example, to maintain quality, the National Council for Teacher Education (NCTE) is mandated with inspecting a sample of existing recognized teacher education institutes annually. But it is able to visit only a small number of institutes every year; in 2011 it inspected 168 out of 14704 institutes.[14] Members on the inspection panel may not have the expertise to do the task. This has led to poor quality of inspections and inaccurate decisions by NCTE.

As an example, the Government of Haryana wrote to the National Council for Teacher Education (NCTE) in 2007 to refrain from granting permission to new teacher education institutes as there was no need for them; yet, NCTE granted permission to 114 new colleges in 2009-10, 37 in 2010-11, 19 in 2012-13 and 19 in 2014-15.[15]

Furthermore, inspections are announced in advance to the institutes. When there is an inspection, institutes allegedly put up a show of faculty and students. There have been instances of recognition granted by NCTE in the face of false or fabricated documents and incorrect information submitted by these institutes.[16] There are also numerous news reports that degrees in education can be bought.[17]

In addition to capacity constraints, regulations have been difficult to enforce because several privately run teacher training institutes are owned by powerful vested interests. There is growing concern that most of these institutes are diploma mills driven solely to make profits, where qualifications can be bought easily. This makes implementing the recommendations of the Justice J.S. Verma Commision and cleaning up the pre-service education landscape difficult.

NCTE, under its former Chairperson, Santhosh Mathew, had initiated a process to enforce strict quality control in the system, using third-party auditors. But this process hit roadblocks, which is unsurprising given that many of these institutes were being operated for profit by powerful interests versus qualifying a cadre of motivated teacher-applicants.

If India is serious about improving student learning levels, it must ensure teachers are adequately prepared before they enter the classroom. The Verma Commission report already tells us what needs to be done. But much greater political will is needed to implement these recommendations, including addressing the capacity constraints required to implement the recommendations meaningfully.

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Manufacturing crisis: the business of learning

ANITA RAMPAL

A decade back, many of us had begun working on the National Curriculum Framework (NCERT, 2005), the primary syllabi and textbooks, with a new focus on understanding young learners, their contexts, aspirations and agency and how they make meaning and construct knowledge, in vastly disparate environments. Bringing together diverse grounded experiences with children, groups of academics, educationists, teachers and artists had departed from the conventional ‘transmission’ approach, where syllabi ‘topics’ are meant to be ‘delivered’ by the system and ‘received’ by students, to promote learning through a pedagogy of empathy.[1]

The same year the EFA Global Monitoring Report had dwelt on the theme of quality of learning and strongly warned that the contrary discourse on ‘efficiency’, of ‘inputs-outcomes’, and the application of production function analysis to education is ‘hazardous’ and diverts attention from classroom practices that have the strongest association with learning and achievement.

Acknowledging the agency of learners and the nature of the environment that motivates their participation in the process of learning, it elaborated that the production function computed by economists for, say, a fence, describes the maximum output (the fence) that may be obtained using different combinations of the inputs – e.g. nails, tools, some planks of wood and some days of labour. ‘But the main difficulty with representing education as a production process is that some of its inputs and all of its outcomes are embodied in pupils, who have their own autonomous behaviour. Planks of wood cannot decide that they do not want to be assembled, avoid coming to the construction site or refuse to interact with construction workers.’[2]

Significantly, in the decade that followed, there has been a dramatic shift towards an economist dominated discourse on education. Planks not pupils are now being sought to be managed, sorted and segregated early, based on their perceived ‘ability’ to contribute to the global economy of ‘knowledge’ or ‘skills’, while showcasing or shaming different nations and governments, in a global race to measure productivity. Ironically, international agencies such as UNESCO or UNICEF, with a humanist mandate for children’s rights and equitable education, have been marginalized and outfinanced. Education policy is assertively governed through the ‘soft political power’[3] of organizations such as OECD supported by funding conglomerates, international banks and large corporates.

The PISA (Programme for International Student Assessment) by OECD, promoting global competitive free market policies, has played a major role in creating ‘PISA shocks’, crisis and alarm, where countries have been coerced into impulsively changing curricula to suit its demands. Ironically, the OECD advice given to Finland was completely contrary to what the country had diligently done to improve learning environments and the performance of all its students, rendering it a PISA winner; similarly Norway was advised that its schools could perform ‘better’ by cutting high spending, closing smaller schools, increasing class size, introducing more testing, publicly declaring test results at school (and teacher) level, and tying teacher payments to students’ results.[4]

The economic priority of OECD's global 'learning' agenda is clear from its claims 'to evaluate the quality, equity and efficiency of school systems in some 70 countries that, together, make up nine-tenths of the world economy.'[5] However, countries showing the highest ranks in international tests in scientific and mathematical literacy are worried because their high scores correlate with pupils showing greatest disinterest in those subjects.

An Open Letter to the Director OECD written by leading academics across the world in 2014, called for a stop to PISA, which 'by emphasizing a narrow range of measurable aspects of education... takes attention away from the less measurable or immeasurable educational objectives... thereby dangerously narrowing our collective imagination regarding what education is and ought to be about.'[6] 'OECD has entered into alliances with multinational for-profit companies, which stand to gain financially from any deficits – real or perceived – unearthed by Pisa.' Incidentally, a close partner is Pearson, which declared itself as 'the world's leading learning company' that won the tender to develop the 2018 PISA Framework.[7] It is also the owner of the *Financial Times*, *The Economist*, Allyn & Bacon, and Prentice Hall, and the largest company involved in testing students and teachers, with related publications and education programmes, operating in more than 80 countries.

The Open Letter went on to state that, 'most importantly: the new Pisa regime... harms our children and impoverishes our classrooms, as it inevitably involves more and longer batteries of multiple-choice testing, more scripted "vendor"-made lessons, and less autonomy for teachers... These developments are in overt conflict with widely accepted principles of good educational and democratic practice.'[8]

Despite these explicit warnings, OECD has enlarged its business model, with a new test for developing countries, and the Indian government has announced it is entering the fray. There seems to have been little 'learning' from India's last misadventure in 2009, when despite MHRD having declined, the World Bank promoted two of our educationally high provisioning states, Tamil Nadu and Himachal Pradesh, into the test. The result was predictable: the media went hysterical, its panellists from Pratham, J-PAL and allied organizations dwelt on the imminent crisis of India coming second last to Kyrgyzstan.

A cursory glance at the test items in science or mathematics would show that teaching even in our well resourced schools does not happen in a manner where students (or their teachers) learn to think through such open-ended questions. My Masters' students have regularly analysed PISA items and compared those to our class 10 board examinations, to reflect on how our secondary syllabi, teaching-learning classroom processes, textbooks and assessments have indeed a long way to go before we can begin to consider subjecting students to such challenging and high stakes testing.

This global manufacture of a learning crisis has several layers of interests, from consultancies, business enterprises for teaching, tuitions and testing, use of information and communication technologies, increasingly for surveillance and monitoring of teaching and learning. Interestingly, while it involves extending financial loans and aid to poor and poorly performing governments for public education, at the same time it exhorts global corporate players to step into the billion dollar industry of 'low cost private schools' for the poor.

The concern for a majority of our children getting poor quality education, even when the country was doing well economically, had galvanized the struggle for The

Right to Education (RTE) Act 2010. It was meant to ensure equitably resourced classrooms (with qualified and enabled teachers, libraries, materials for experiments, etc.) for all our pupils at the elementary level, to participate in the learning process with confidence, through exploration, discussion and discovery, building a good foundation for subsequent secondary schooling. However, organizations advocating for more testing within the input-output model have continuously attacked the RTE, claiming that learning levels have declined since its implementation.

This, however, is questionable, as also shown by a recent study by researchers at the Indian Institute of Management Ahmedabad.[9] Levels had been falling before 2010 and though arrested in some states after the RTE, did not significantly improve as often happens when enrolments go up sharply while the system fails to make provision for adequate teachers, good classrooms and learning resources. More significantly, the legal authority for the RTE, the National Commission for the Protection of Child Rights, has recently noted that the continuous and comprehensive assessment of the learning of each child as had been mandated by the RTE, to track their progress all year round, was conducted in a completely flawed manner by the Central Board of Secondary Education, which also had no *locus standi* to be doing it for elementary schools. Lack of proper implementation of the RTE, thus actually breached children's fundamental right to participate in school, learn through nurturing learning environments, and be assessed through an authentic pattern of tracking and supporting their development.

The RTE Act (for children in classes 1-8) has two separate sections – 16, which says 'no child shall be held back or expelled from school till the completion of elementary education'; and 30(1), which says 'no child shall be required to pass any board exam till completion of elementary education'. The RTE rightly differentiates between a school examination and a centralized board examination – the first needs to be close to the learning context, rooted in the child's culture, language and environment. The board examination, on the other hand, is set at a distance by people who may be far removed from the social context of the child and also may not know what teaching-learning has happened in the school. Moreover, board exams are generally competitive and, therefore, lead to more stress even among high performing students.

Learning theories and researches show that students (even at higher levels) perform best in a non-competitive, non-threatening assessment or examination environment. As soon as there are high stakes involved, or in a 'selection' where large numbers compete for a few seats or positions, the performance of even the best drops.

Examination and curricular reforms in Kerala (initiated during the District Primary Education Programme) had shown that in districts where question papers were set by teachers in school clusters through creative discussion, and each item was considered a 'learning activity' conducted in a non-threatening manner, children's performance across a wide range of abilities was much better than in districts following a conventional pattern.[10] However, even after the RTE, school exams have remained regressive, focusing on rote memory and narrow skills of information recall, with no attempt to promote critical thinking, original expression, innovation or creativity. Detaining students based on such examinations is in itself questionable.

The official MHRD note accompanying the RTE had stated the rationale for Section 16 as: 'The "no detention" provision is made because examinations are often used for eliminating children who obtain poor marks. Once declared "fail", children either repeat grade or leave the school altogether. Compelling a child to repeat a class is demotivating and discouraging. Repeating a class does not give the child any special resources to deal with the same syllabus requirements for yet another year.'

It went on to assert that each child has the same potential for learning; a 'slow' learner or a 'failed' child is not because of any inherent drawback in the child, but most often reflects the inadequacy of the learning environment and the delivery system to help the child realize his/her potential, meaning thereby *that the failure is of the system, rather than of the child.* This requires addressing the improvement of the quality of the system rather than punishing the child through detention. There is no study or research that suggests that the quality of learning of the child improves if the child is failed. In fact, more often than not, the child abandons school/learning altogether.

The most disadvantaged first learners, who are neglected, taunted and labelled as 'slow', from whom teachers have least expectations, and whose knowledge and 'lifeworlds' do not normally get reflected in the school curriculum, are those who are most vulnerable to be detained, and pushed out. Research tells us that 'motivation' and self-esteem is crucial for learning, and teachers who are 'warm demanders', challenging and nurturing students, manage to get all children to learn well, though at different paces. Giving them time and support is crucial, while acknowledging what they know and can do well. Stigmatizing and detaining only lowers the self-image and further demotivates them.

Students learn not as isolated beings (certainly not staring at blackboards, if those exist, or textbooks if they have any) but from interaction with their peers, in mixed ability groups, actively involved in thinking, discussing, observing, exploring, presenting, and so on. Do our classrooms seem remotely capable of serving as 'learning spaces'? Even our ostensibly better schools are only teaching rules to be followed, algorithms and answers to be copied and reproduced.

More worryingly, the Delhi government has gone a step further in using the learning crisis to discriminate and segregate children from class 1 onwards, into separate sections on the basis of their so called 'ability'. It uses differentiated syllabi and tests for children assigned labels such as '*pratibha*' (talent) or '*nishthha*' (commitment). It has also called for revoking the RTE to begin detaining them from class 3; while a Central Advisory Board Committee has recommended that 'skill' based vocational education should be offered as early as class 3! Promoting the business of surveillance, the Delhi government allocated over a thousand crore just for CCTVs and, playing on parents' fears, is enticing them into the 'technological fix' of live streaming on their phones, thus not only shifting some responsibility onto them for ensuring the safety of their own children, but also completely invading the pedagogical space of the classroom.

The alarming state of 'biocontrol' today involves not just monitoring teaching and learning through cameras and live streaming, but also more invasively through digital tracking devices on pupils' bodies (including electrodes and straps to scan the brain, eyes or skin activity), or even implants inside them, through enormous corporate funding. In 2012 the Gates Foundation in the U.S. had given 1.1 million dollars to a university for the use of 'Galvanic' skin bracelets, strapped onto children in classrooms. This research grant was part of the Gates Foundation's 'Measuring Effective Teachers Project', on using biometric technology to monitor emotional and cognitive response to specific stimuli, in order to measure children's engagement while learning.[11]

In fact, millions of dollars had already been spent on evaluating teachers' quality of teaching, through standardized tests, questionable video-taping of their lessons, with contested results. Protests had raised questions about the efficacy and ethical dimensions of this research, especially when hundreds of American schools had been facing essential shortages, including funds for teachers' salaries or the school electricity bills. Students and teachers alike had found such snooping devices repugnant and there were parodies on how they would raise their ratings on the 'learning' bracelet by getting children's pulses or neurons racing.

Children sense these injustices done to them. They have all along wondered why they face discrimination or controls in schools, private or government, Hindi or English medium, in sections for 'slow' or 'talented' learners, for the rich or poor… Pupils' voices (translated from Hindi) in the following discussion clearly show how motivation and agency, determined to a large extent by the ethos of school, play an important role in either their autonomous decision to participate in learning or to give up. They invoke a commonsensical understanding of quality and social justice and the potential of a good learning environment, to declare that if there is discrimination even in schools then what is the point of going there. These are pupils, not planks, speaking, feeling, acting, demanding respect, their right to learn and 'together help each other move ahead':[12]

Pupil 1: Now see, this is a government school, and *not one* child of a doctor or engineer studies here! (continues emphatically) When the child of a doctor and the child of a junk collector come to the same school, they study and play together and get to understand each other better. They will learn useful things from each other's lives and become friends. Then there should be no problem. Yes, at first it may take some time, but if teachers take care these children can become friends soon enough.

Pupil 2: But ma'am here things happen in just the opposite way. I mean, if a child is very poor the teachers themselves make fun of him. See this poor chap (pointing to Babloo), how they all deal with him! Not one teacher talks to him properly. They even make him sit separately saying, 'You are in any case not going to understand anything'.

Babloo: (in a muted, sad voice, almost struggling to speak) Ma'am, even poor children want to study, want to move ahead and become someone. They may not have the facilities but they are very hard working. They need just a little love and encouragement.

Pupil 1: ... if teachers were to explain well then all children will learn to love each other more, and *will together help each other move ahead*. In our class all teachers talk so lovingly with Monu, they also increase his marks in the exams because he is from a slightly better off home. But they all behave very rudely with our Babloo. This poor fellow was telling us once that his mummy is always ill; his papa gets work only on some days, not on most days. He eats *roti* (traditional bread) sometimes only with salt, or with an onion. And then madam talks to him like this: 'Your mummy and papa do not take care of you. You land up in school as a "*chooda*" (an abusive word implying a dirty sweeper).'

Footnotes:

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Is there a learning crisis in our schools?

DISHA NAWANI

IN my understanding, the term crisis indicates a moment of extreme consequence or calamity when crucial steps need to be immediately taken in order to prevent a potential catastrophe. In the Indian educational context for some time now, there has been discontent expressed relating to both the 'nature/quality' of learning conceived of/transacted in schools and the sheer 'presence/absence' of it among school-children.

While the concern that we need to seriously reflect on the kind of learning imparted in schools is a relatively old one, the concern that most schools are not even enabling children to acquire basic literacy and numeracy skills to their students is relatively new, and one that has grabbed a lot of media attention. Compared to the former, the latter presents itself as a crisis that needs to be addressed with utmost urgency. Despite the knowledge that there exist multiple conceptions of learning, since this is projected as the least common denominator/skill that all children must learn, its reported absence among schoolchildren acquires the status of a near catastrophe.

This paper highlights two kinds of learning crises in the Indian educational context, both of which have received either periodic or consistent attention but have widely differing implications for curricular, pedagogic and assessment related matters in schools, the importance of teachers – their education and service conditions and role of public institutions and private partners in enabling learning. This paper is divided into four sections. Issues ranging from crisis of not knowing, stratified schools and crisis pertaining to an absence of meaningful learning are discussed in these sections. The last part of the paper concludes with a brief discussion on possible future steps.

Let us briefly first discuss the learning crisis which has almost become a matter of national shame and ridicule. This type of learning is defined by different assessment surveys and has largely emerged from the findings of the Annual Status of Education Report (ASER) since 2005, which has been shouting aloud that rural children going to schools are not even equipped with age-appropriate basic skills which are absolutely essential for any kind of formal learning to take place. For unpacking this concern, we need to have clarity on the following questions: (*i*) What is the nature of learning that is being talked about? (*ii*) How is that learning measured? (*iii*) How does one factor in differences among individual children/schools? (*iv*) How do we make sense of these reports?

To briefly answer these questions, the nature of learning assessed in these surveys is limited to a minimal knowledge of literacy and numeracy skills which have a bearing on the age of the child and grade that she is studying in. Learning is measured by simple oral tools – questions asked which need to be answered by the respondents. Such tests do recognize context related variations among the children but do not give them sufficient importance. The more fundamental question perhaps is, how one should view the results of these surveys.

* Any kind of educational measurement is only an approximation of particular kinds of abilities. Yardsticks applied to measurement of physical features, such as height or weight cannot be applied equally to the study of mental processes like thinking, abilities and so on. At most, one can attempt to *infer* cognitive processes

based on some evidence but not claim to provide accurate, unassailable accounts of learning.

* All assessment tools have some limitations and even if data drawn from them is reliable and valid, they provide only a partial picture with a certain fixed idea of learning. Each such tool has an underlying theory of learning and attempts to capture a certain kind of evidence based on that theory. While for some, the outcomes alone would be a fair indicator of learning, for others, processes by which one arrives at that outcome would perhaps be of greater value.

* Such assessments can measure only those abilities which are amenable to objective manifestation and standardized quantification. Added to this is the fact that such data is collected within a specified time period and less emphasis is given to either rapport building or allowing the respondents to formulate their answers at ease.

* The visibility, hype and linkages of these surveys with the notion of 'quality' education and 'accountability' puts enormous pressure on the teachers to 'teach to the test' and on schools in turn to produce results. This is because by showing/exposing students' learning, they make a comment on not only students but teachers, schools and education systems at large. This results in the meaning of learning getting reduced to what will be asked in these tests and quality equated with students' performance scores in them. Even if these basic skills are absolutely integral to a child's learning, they capture a limited and fixed idea of learning which violates the fundamental principles of viewing the teaching-learning process as an emancipatory exercise where students and teachers jointly make sense of the world.

* Unfortunately, discussions around large-scale assessments mostly revolve around the mechanics of testing, sampling design and efficacy of the tools used and not so much on its implications on learning, pedagogic interactions in classrooms, role of teachers and importance of prior knowledge and experiences of the child.[1]

These tests focus on the non-learning of rural children and have been particularly candid about those studying in government schools. Besides the disputed methodology of such tests, the assumptions underlying them regarding the 'superior' learning of private versus public institutions, and the positioning of teachers in these institutions is highly problematic. (*i*) The 'finding' that private schoolchildren perform better than those studying in government schools underplays the extremely constrained and challenging backgrounds that the latter come from. (*ii*) The 'proposed' solution – that private schools should be allowed to flourish without being subjected to the heavy-handed regulations proposed in RTE Act is likely to be misused by the profit making private lobby which promotes private education of the poor as a viable enterprise. (*iii*) Overall there is an undermining of the welfare oriented public institutions without appreciating that their 'alleged' dysfunctionality cannot be resolved by eliminating them. (*iv*) The faith in the competence and motivation of under-trained, underpaid and contractual teachers vis-a-vis secure government teachers is misplaced and undermines the integrity of the teaching profession. (*v*) More importantly, the implications of such tests on schooling processes are such that they tend to reduce the meaning of learning to a minimum core where curriculum, pedagogy and assessment are all geared towards ensuring the acquisition of basic skills by students.

Having said that, even if a few school going children are unable to pick up basic skills which form the foundation of their formal learning, then it is certainly a matter of concern. However, rather than getting into a blame game and advancing simplistic cause and effect relationships, one must systematically try and examine their reasons.

Before describing the other kind of learning crisis, it is important to recapitulate the nature of fit between it and Indian society for there is a direct relationship between the learning crisis being talked about and the social-cultural contexts of children, whose learning is in question.

In line with our hierarchical social structure, the Indian school system is also visibly divided, leading to a fragmentation of experiences of children studying in different schools. Broadly speaking, our schools fall into a four-tiered system. The top tier constitutes the elite schools which include the exclusive unaided private schools, with affiliation to international boards etc. Admission to them is restricted due to their steep fee structure. The next layer comprises of government central schools and 'good' quality private aided/unaided schools. Entry to them again is either dictated by some mandatory requirement on part of the child/parent or their capacity to shell out reasonable money as fee. The third tier includes private schools, aided or unaided, of average or indifferent quality. Finally, there is a stratum of provincial or regional government/local body schools and low cost private schools that cater to poorer segments. Just like the *varna* system, this is not a neat but a very wide categorization and has several intermittent layers of school type within it.

To reiterate, school systems – be they public or private – are internally hierarchized and there is a fairly straightforward direct relationship between the socio-economic backgrounds of children and the kind of schools they have access to. There are several parameters on which these schools differ from each other, ranging from infrastructure to quality of teachers employed in them. On one hand, there are schools which have adequate resources to create a desirable learning environment for children and on the other, there are schools without proper classrooms, teachers and even toilets, leave alone playgrounds and laboratories. The schooling experiences of poor and disadvantaged children is constrained in several respects across schools. Challenging home environments and minimal pedagogic environment/curricular resource support in schools impedes their learning.

This means that blanket statements like there is a learning crisis across all Indian schools are not necessarily true. The reason for this is because the complexities of the Indian social fabric resonates in equal if not more virulent forms in its schools. Consonance between the two provides an ideal breeding ground for social inequalities to not just get reproduced but legitimized through its schools.

The privileged remain cocooned in their protected environment and are nowhere close to the crisis of not learning which is documented in the above mentioned learning surveys. It is another matter that children across schools – from both advantaged and disadvantaged backgrounds have access to a weak foundation of learning and therefore confuse memorizing inert content with learning. The next section draws our attention to the real crisis in learning which needs to be addressed with earnestness.

Learning is a ubiquitous phenomenon which does not happen only within the four walls of a school. Interestingly enough, schools as especially established spaces with a monopoly over formal learning make a sharp divide between learning inside and outside school. Besides serving an instrumental value for certified learning, this dichotomy also exacerbates burden of school learning for the child by making it an insular, sterilized experience.

The divide in learning between these two spaces of school and home/community and the need to bridge the two is also another persistent concern which has been periodically highlighted by several commissions, academic bodies and researchers, but is presumably regarded as being innocuous since it does not threaten the very existence/objective of schools, and is therefore, not seen as a crisis. So much so that we even live with non-access, limited access and defunct access of some children to state funded public education as a normal feature of our society. It took almost 60 years from 1950, the year independent India adopted its Constitution to 2009, when the RTE Act was finally enacted, to question this.

The meaning of learning across most Indian schools affiliated to both central examination boards or state boards is restrictive and confined to textbooks prescribed by them. In this sense, the privileged and underprivileged child is perhaps in the same boat, except that the former has more support to counter the menace of textbooks/rote learning. Studies also reveal an absence of higher order thinking skills even in students studying in private, elite schools.[2]

It has been seen that the knowledge that children acquire in schools bears little relationship to their experiences, perspectives and world views. The image of an ideal learner is one who is obedient and quiet. It is evident that no singular unified curriculum can possibly represent the diverse childhoods in India and that's the reason for emphasizing the need of a contextualized curriculum and teaching-learning resources. Time and again, it has been reiterated that children's voices and experiences need to be recognized in the classrooms so that they, along with their teachers can participate in construction of knowledge.

Learning in most schools is encapsulated in a narrow and focused fashion in curricula, syllabi and prescribed textbooks. More importantly, it is believed that textbooks contain all that is worth teaching, knowing, assessing and certifying. Textbooks are an integral, almost indispensable reality of the Indian classroom. The curriculum framework is a fairly new idea in the Indian education system. While syllabi for different subjects is readily available to teachers and students, often even that is not referred to in the classrooms. It is the prescribed textbooks, which supposedly *contain* all that there is to be learnt. Besides the fact that textbooks are specially crafted in accordance with the syllabus, almost all formal modes of assessment, be they tests or exams are based on them, thus gracing them with infinite importance and legitimacy. Teachers teach from textbooks, using the same examples, often exclusively relying on them to create boundaries between curricular and co-curricular knowledge.

The problem is not that textbooks are used for learning but that they are most often the *only* resource used to the exclusion of other resources and that they are essentially memorized and reproduced verbatim in exams. While the pedagogy in a majority of schools across the board is textbook driven, teachers in some schools do use additional resources/techniques to teach and those students who are equipped with desirable cultural and social capital are probably better placed to show an enhanced understanding of the knowledge imparted in schools.

It is also a fact that schools through their curricular resources, primarily textbooks and pedagogic processes and interactions with students, commit symbolic violence on those who are different in some respect – smaller in number/from a distant habitation/disadvantaged in terms of gender, religion, ethnicity by either under – or misrepresenting them, thus leading to a prejudiced public perceptions/attitudes towards them. This dehumanizes both the perceived and those who perceive these others in a certain way.

Another 'evil' repeatedly pointed out concerns the form and objective of assessment practices in our schools. One of the misconceptions around which our education system is built is that: 'Learning in schools gets established only after it is assessed, measured and certified in a formal standardized manner.' This approach, where assessment is seen as a tool to test learning is seeped deep into the psyche of formal schooling structures. Indian schools have revelled in the glory of assessing their students through externally conducted public examinations, which was a legacy of the colonial education system.

These one-off terminal exams put excessive stress on students, tested their ability to memorize prescribed content and appropriately rewarded/penalized (promoted/detained) the students. A system which respected the anonymity of both the teacher and taught, which was uniform and seemingly impartial, was posited as being fair and a suitable mechanism for scuttling the aspirations of millions by failing those who were unable to meet the requisite standards.

Several committees/policy documents have pointed out the idiosyncrasies of such a system as violating both the idea of a meaningful learning and children's right to learn with dignity and suggested alternative school based reforms. Central to this was the proposal of a school based continuous and comprehensive system of assessment which broadened the scope of learning and recognized that teachers were best placed to make such judgments and also, that students were competent to manifest their learning in multiple ways.

Aligned with this provision, which found place in the RTE Act, was another provision, which mandated that no child until completion of elementary schooling would be detained in the same class even upon failing. This provision did not go down well with our policy makers, parents and teachers, who believed that this promoted a lackadaisical attitude among children, who now in a fear-free environment became carefree and stopped learning. Ridiculous as it sounds, this is true – the policy was promptly reversed and board exams reintroduced as early as class V.

To recapitulate, the crisis in learning, in my view is at two levels. The first, curricular and pedagogic, addresses the 'burden' of school students. A report, aptly titled, 'Learning Without Burden'[3] defined the burden as one of 'incomprehensibility' – a system where there was no joy in learning and where, 'a lot was taught but little was learnt or understood'. Not that this malaise was not known earlier, but it was presented coherently for the first time. The points of intervention suggested by it were largely related to curricular, pedagogic and assessment dimensions of school education. It also laid emphasis on fixing the structural deficits in schools and ensuring robust teacher education programmes. Besides culminating in a National Curricular Framework in 2005,[4] followed by new textbooks prepared by the National Council of Education Research and Training (NCERT), this concern was not really viewed as a crisis demanding immediate attention.

The second focuses on the 'learning deficit' as indicated by a range of assessment surveys which basically measure the acquisition of minimal skills by children in schools. Curriculum, pedagogy and assessment processes in schools were not its concerns. The two central arguments of these surveys seem to be that (*i*) since government teachers are secure in their jobs and overpaid, they manifest less accountability towards students, and (*ii*) that government schools are failing miserably in their job. Unsurprisingly therefore, their suggestions are drawn from these findings – employment of contractual teachers (presumably more accountable) and reliance on low cost private schools to teach children of the poor.

It would be naive on my part to give new suggestions, other than those which have already been offered in the past. It is ironical that while we have been able to identify and acknowledge problems in our education system and also offer reasonable ways to deal with them, we still have not been able to suitably resolve them.

For instance, it has been reiterated *ad nauseum* that a singular, standardized resource like the textbook in a large heterogeneous society like ours cannot represent all our children's lives and consequently a majority of them will always feel alienated and left out. Moreover, it will lead to only a particular kind of learning and likely jeopardize the entire meaning of learning. While textbooks have tremendous advantages, they cannot possibly encapsulate all that the child needs to know and learn. Moreover, children learn in different ways and it is important that they are provided with a range of resources that cater to their needs, learning styles and contexts. Textbook learning is integrally linked to learning by rote, which has an important place but should at no cost be confused with any meaningful learning. It has also been suggested that textbooks should be vetted for both their content and messages – overt and covert – that they convey.

In this context, mention must be made of the National Curricular Framework (NCF), 2005. It was a fairly significant document which explicitly linked the problem of 'dropouts and issues of inequality and quality in education' to the 'nature of learning children experienced in schools'. It upfront recognized the importance of a contextualized, vibrant and meaningful curriculum, which not only acknowledges the agency of the child but also pleads for a pedagogy and environment where children can learn without fear.

The NCF reiterated the failure of our curriculum and schools to evoke the interest of children and retain them in schools. It stated that the issue of quality needs to be centrally linked with the educational experience that children have in schools and efforts should be made in the direction of ensuring an all-round development of the children, building on their knowledge/experiences and making them learn in a positive and fear free environment. There are several such solutions suggested pertaining to teacher education and structural prerequisites to ensure meaningful learning in schools.

Then there are also solutions proposed towards a shift in institutional responsibilities. The low cost private school lobby has been making its presence felt rather aggressively, claiming that they are doing a far better job than the government schools, and hence they should be given a greater responsibility in the schooling of primary children. Their central argument is that such budget schools are economically more viable and learning of children studying in them is better than those studying in free government schools. This, I believe, is a dangerous trend in a developing society like ours and has implications for the education of those poor, who can only access free government schools.

Research has shown that low cost private schools which are being offered as an alternative come at a cost and offer education of a dubious quality (where meaning of education is restricted to minimum manifest learning) under constrained circumstances (space and teacher quality). Rather than throwing the baby with the bathwater, we need to work with government schools (not that all public schools are bad, and not all private, good) to make them better, and not disown and abandon them.

The larger question is that whatever the schools are doing in the name of learning is clearly not enough – whether this means ensuring an enabling environment in which children make sense of the world around them and their own locations in relation to it or even acquiring basic skills. Almost every aspect of possible educational reform suggested to address the malaise plaguing our educational system, if implemented, will have a meaningful but limited role. If we want to ensure meaningful, equitable and quality education experiences for all our children, then we will need to restructure our schooling system which at the moment is in complete sync with our stratified social structure, and perhaps de-link 'marks obtained' in exams with admissions to higher institutions and jobs. Unless and until this congruence is disrupted, no fundamental change will be achieved and different students will continue to have differentiated learning experiences in varied schools that they study in.

Footnotes:

1. D. Nawani, 'Assessing ASER 2017: Reading Between the Lines', *Economic and Political Weekly* 53(8), 2018, pp. 14-17.

2. Schools affiliated to international boards are not within the purview of this commentary.

3. Government of India, Learning Without Burden: Report of the National Advisory Committee Appointed by the Ministry of Human Resource Development. MHRD, Department of Education, New Delhi, 1993.

4. MHRD, National Curriculum Framework. NCERT, New Delhi, 2005.

Inclusion and equitable education

H.K. DEWAN

MUCH of the recent writing and policy discussion on learning seems to reflect a view that our public provisioning of education is dysfunctional and that teachers, particularly in public schools, despite receiving reasonable remuneration, do not teach satisfactorily. This has led to an enhanced focus on tracking every movement/shift in learning. The methods of measurement have become more sophisticated, and the resources and personnel involved have increased manifold in an effort to ensure better learning outcomes. This apparently is the emerging consensus about the way forward to ensure that all children learn what we think they should learn. It is, therefore, crucial that before this new orthodoxy is accepted as *fait accompli*, those involved with education re-examine the underlying beliefs and principles that have gone into this construction.

Over the past few decades we have witnessed many new schemes, initially in primary and elementary and, subsequently, secondary education. Additionally, new state run teacher training institutions have been set up to improve the quality of teacher education. Unfortunately, since these indifferently executed efforts have expectedly not resulted in any discernible improvement, the conclusion is to cut back on all this basic investment and instead focus on monitoring, tracking and measuring outcomes along with such interventions that could be rapidly implemented so as to ensure a better learning society.

Consequently we have new initiatives to come up with 'materials' that a child can use herself with the role of the teacher reduced to a facilitator, a far cry from the earlier belief about teachers as guides and exemplars. Some even propose that learning can be done through use of mobile/internet technology replacing a teacher and that we no longer need to fund schools to levels suggested. Others argue that the problem is primarily with public provisioned schools and that an increased reliance on the private sector, for instance through public-private partnerships, is the way forward.

Given the vast number of groups working at both the Union and state levels to improve the learning standards of our children, it is time to take a close look at their propositions. Four key aspects that we think need examination are: (*a*) What is to be learnt and why? (*b*) What does equitable opportunity mean? (*c*) What does it mean to be a teacher? and (*d*) Have we been governing and administering education well?

The current push is towards ensuring greater administrative control over the government teacher, accentuated by the belief that they are often absent, play truant and, when in school, do not teach well; that they do not know the subjects or how to teach them and despite all efforts are demotivated and hence become objects of ridicule and derision, 'unworthy' of being respected as teachers. The implication is that little purpose is served in trying to empower or motivate them. And with confidence it is claimed that the methods and materials suggested, if followed properly, would lead to improved learning in schools. Intriguingly, most groups that work in this space are convinced that the programmes and package they have developed or are developing would be the most appropriate. Ironically, most of these efforts fail to recognize that a desire to express ideas, think and analyse alternatives is the key to human agency and if the school system has to function well then teachers too need the space and the ability to think independently and make use of their own ideas.

This desire to promote 'the package' combines well with the desire of the administrator to control the day-to-day activity of the teacher. Current 'educational administrators' feel most satisfied when all teachers and schools under their control, irrespective of their specific context, do the same thing at any given time. This orientation meshes well with those constructing and advancing measurement and monitoring strategies and those who create packages of materials and methods for children and teachers to follow. The justification for this rests on the argument of inadequate resources available for expanding schooling, the lack of capable, well informed and motivated persons to teach, and the belief that while education involves a wider set of elements, the relevant attribute is the ability to read alphabets, to count and read numerals. There is also talk of 'HOTS' (the so-called) higher order thinking skills to be developed.

One common trait is the 'mis'-use of terms (child centred, facilitation not teaching, activity and self-learning, developing skills and competencies etc.) by reinterpreting them and altering their original meaning unrecognizably. There is a need to examine closely these terms and spell out what constitutes learning, its purpose, how human beings learn and implications for the school and society. We must also unpack the marked impatience with any suggestion to include everyone, of democratizing learning and its opportunities and possibilities.

The urge is to ensure that the investment of resources does not continue to endlessly increase results in rapid shifts in strategies and programmes of action. The goals and purposes in the process are also narrowed. There is deep scepticism and cynicism about the ability of the poor, socially and culturally different children to learn. It appears that those charged with the responsibility of ensuring equal participation in education have little faith in the ability of disadvantaged children to learn and compete with their middle class compatriots.

The current drive towards stricter and narrower processes of measuring the so-called expected levels achieved, the content of learning and the thrust towards monitoring of teachers without granting them the agency needs to be examined through this lens. The patience to engage in a dialogue on these concerns is wearing thin. What is important to be taught and learnt and how it must be learnt is increasingly being defined and controlled by those who have no direct experience of working with teachers or children and often lack a sound theoretical understanding. The entire effort seems to be driven, therefore, by catch phrases and misconceived principles. It is almost as if everyone other than the teacher knows better than the teacher. The dominant attitude among administrators is, at best, benevolently disdainful of teachers, the children and their parents. They are not seen as individuals but numbers to be slotted into categories.

The system has forsaken education as a transformative process requiring dialogue, scaffolding and empathy with those involved in the teaching-learning enterprise. Believing that the highest ranked bureaucrat is the most responsible, most committed, most knowledgeable and the wisest and, therefore, will present the best strategy, the system faces the crisis of advancing something that many of those actually involved do not believe in or understand. The irony is the 'wise' administrator is the wisest only till the next wise administrator replaces him/her and declares that all that was being done till then needs to be modified and altered. Rapid change of plans and strategies leave people in the system, particularly the teachers, bewildered and cynical. Even before the system is in a position to respond and the teachers begin to understand the proposed ideas, they are presented with a new direction that they need to follow.

The quest to universalize education is relatively recent. Notwithstanding the Constituent Assembly discussions, the promise of equitable opportunity was virtually forgotten. True, some policy documents and commission reports stressed the need for increased budgetary allocations, alongside ensuring greater respect, autonomy, time and opportunity for personal growth and development of all teachers but none of that ever happened. The proposed common school system, for instance, was given a quiet burial before birth.

The 1986 policy, while repeating several ideas from the 1968 policy document also signalled some subtle and some fairly obvious shifts. The entire enterprise of education, the role of the state, the definition of the teacher, the nature and purpose of education and the construction of a citizen underwent significant changes. The individual citizen, for instance, became a human resource, and from a constitutive element of the nation became someone to be shaped as a resource for the nation, a shift aptly captured in the renaming of the Ministry of Education as Ministry of Human Resource Development. The teachers could now be paid differential salaries and subjected to closer monitoring; science was reduced to technology, detracting from its role as a source of reason and rationality. The 'programme of action' not only discarded the common school system, it legitimized a differential school system, including in the public sector. It also enhanced the space and role for private provisioning. The logic was that a poor country just cannot afford continually enhanced allocations on public schooling. A combination of centrally sponsored schemes and soft loans from the World Bank and other grants were seen as insufficient.

The gap between intent and provision has been apparent in the limited budgetary allocations. Operation Black Board and subsequent similar initiatives promoted a minimum rather than essential school infrastructure. Non-Formal Centres (NFEs) were proposed for the masses alongside the well endowed schools (Kendriya Vidyalaya, Navodaya Vidyalaya) for the bright and middle class children. These schools, however, also soon lost out to better endowed private schools as 'elite' children shifted out in large numbers. The policy thus speaks in two voices – equitable education and the need for nurturing talent. We continued to pay lip service to the idea of universal primary education but encouraged and promoted non-formal education centres, *shiksha karmi* schools and so on. This leaves one wondering if there was ever a commitment towards equitable education.

On the one hand this has led to a rubbishing of the public education system as wasteful and inefficient; on the other there are complaints about the arbitrariness of private schools in matters of fees and admissions, resulting in a call for regularization. The vocal rich, middle classes and government employees are focused on their own interests – namely regulating fees and ensuring the superior performance of their children. The top-down attempt of the government to set up parent-teacher associations or school management committees remains half-hearted and futile. So, while there is the rhetoric of everyone needs to be educated and that the government is committed to educate everyone, this is not followed up. Unsurprisingly, the conviction to take this forward is lacking, despite the Right to Education Act 2009, National Curriculum Framework 2005 or many other commission reports.

The RTE, for instance, defines the school, quality, teacher and everything else that is embedded in the educational enterprise, but fails to specify how its proposal could be contextualized or how high standards could be achieved. It makes everyone answerable to the appropriate government and local authority without specifying the role of the community, social and academic institutions. The essential criteria spelt out are limited to infrastructure and a few observable parameters, leaving the definition and interpretation of educational terms to the courts. In one stroke the act has made all initiatives aimed at educating excluded groups 'inappropriate' and illegal, leaving no space for alternative models and approaches.

Another much talked about aspect of RTE is age appropriate education. The act (chapter V, section 29), says that curriculum and evaluation will be in conformity with constitutional values and directed towards all-round development to enable children to realize their potential and talent. It emphasizes learning without fear, trauma and stress. Words like child-friendly, child-centred and learning through activity, discovery and exploration abound in the policy. The act also did away with examinations in the elementary stages and introduced CCE – continuous comprehensive evaluation. The government also proclaimed that no child would be denied admission and all children would be placed in age appropriate classes. However, nowhere did the act or subsequent policies spell out how a child would be able to reach the 'appropriate level'. The teacher was expected to somehow find extra time and perform this miracle without any learning support. Hardly any funds were provided for this.

The gap between the rhetoric and the intent is stark! The norms and standards suggested for a school are grossly inadequate when compared to even ordinary private schools. The different norms for government run Kendriya Vidyalayas and Navodaya Vidyalayas also demonstrate the low conviction in equitable educational opportunities.

Section 12(1)(c) of the act says that a private school shall admit *at least* 25% of children from economically and socially weak backgrounds in class I and that the government would reimburse expenditure up to per child cost incurred by the state in its schools. If the intent was to move towards equity and inclusion, then the effort should have been to make schools admit more than 25% children and in other classes too. The regulations do neither. The stipulation 'at least' actually reads as 'at most'; and children already in school or children needing to join age-appropriate classes are excluded. Finally, the mistrust and corruption within the system makes release of funds for these children an arduous task. For the high end private schools, the funds offered are in any case too meagre in comparison with their stated expenditure. The real inclusion of these children in the school and meeting the difference in the cost calculated by the school remain a major challenge. Both these constitute a headache for schools already disinclined to admit such students into their classrooms and indeed into their premises.

The question then is why more comprehensive possibilities like common schools or larger reimbursements to match the expenditure of a particular school cannot be considered. It is essential for the child to not be a forced obligation and receive equal treatment.

The overall pattern is revealed in a reluctance to invest in the general education of poor children. Schools for them are expected to make do with low resources and function at low cost. The disinclination is reflected in views like: (*a*) Since payment to teachers is a serious burden on the exchequer, one must have more private schools. (*b*) Why hire regular teachers when para/contract teachers can suffice? (*c*) Since funds from international agencies as grant or loan cannot be used for salaries at any level, the allocation in the state budgets for schoolteacher salaries is reduced as state governments are expected to share costs incurred on schemes/programme of the central government. Consequently, even as the educational outlays of the central government have not increased in any substantial manner, the expenditures in areas other than staffing have gone up, for example, on infrastructure, materials, ICT and consultants.

The failure to educate, motivate or enthuse the school system has been attributed to disinterest of teachers. The convenient conclusion is that neither are the teachers capable, nor can they be made so and hence the need to invest in teacher-proof materials and methods. Intriguingly, teacher-independent techniques are defended on principles like learner centrality, self-learning, problem solving approach, learning not teaching, facilitation, constructivist learning and so on. The enthusiastic entrepreneurs 'supporting' public education are convinced that since the government is in no position to spend additional money on education, the school should focus on teaching the basic alphabets and numbers.

The new focus on tracking each individual child on specified parameters suggests that education is seen like an industrial process – a belief that since all children are similar, if put through identical processes, the end product too would be similar. The outcomes, therefore, can be tracked, recorded and measured. This proposition rests on the belief that data for diverse children, collated month after month, can be meaningfully studied and analysed and the analysis could initiate change in teachers and the entire school system. Unfortunately, this data analysis, while useful for ranking schools or regions, does not help individual teachers improve. The data can be meaningfully used at the school level *only* if the teacher is both allowed and expected to formulate her own strategy, follow her own instinct, set her own exam paper and judge the responses for herself.

The school when conceived of as a plural secular space can greatly impact the children and their childhood. Leaving aside for a moment the misconstrued, one-dimensional pressure some parents place on their children, the school offers the child a place to express and celebrate her childhood. She has no chores to complete, learns to socialize with other children and, for a brief while, is freed of the responsibility invariably assigned to her when at home. We have never asked children, particularly girls, about what they see in the school and why they go there. What is it that they learn there and how can the place become more welcoming?

We also rarely ask whether the most important element of preparing a citizenry is knowledge of alphabets and numerals? Nor whether children from different backgrounds and circumstances acquire skills in the same manner and at the same pace? Do we even care to consider (like *nai talim* did) the distinction between being literate and being educated? We rarely ponder over the importance of language and experience of children, the value of using local specific materials and (most importantly) the critical role of teachers (and other adults). Clearly, despite the advice of academics and educators and even recent NCERT curricular statements (NCF 2005), the division between 'scholastic' (largely cognitive) and 'co-scholastic' (identified as 'affective') has become sharper. The mainstream push is on acquiring communicative skills (besides familiarity with alphabets and numbers) to create a pliable consumer and market employable worker, contrary to the spirit conveyed in various policy documents.

These processes also alter the meaning of the school as a secular inclusive space to help build a sense of respect for reason, plurality of views and behaviour, and for cooperation. If this is indeed our goal, we need fully engaged children participating in all processes, and teachers who respond positively and constructively to all situations in their school. Equally, the school needs to respect children who attend in spite of all the difficulties. Instead, we now consider children who come to school for the mid-day meal as freeloaders. The teacher ideally should be concerned and encourage them to stay on longer, even if only to play and be with other children, glance at books and get a feel of childhood. Teachers engaging with children inclusively, in a creative and open-ended manner, would keep them at school.

The idea of CCE with the teacher at its centre requires her to deal, as far as possible, with the learning of each child individually, even though the activities are collective. She needs to track the learning of each child and praise and encourage her – instead of constantly pointing out her distance from the an 'artificially' constructed age norm. The manner in which the no-detention policy and CCE have been implemented demonstrates a marked lack of sensitivity for children struggling to stay on in schools. The CCE should encourage teachers to shed the attitude of completing the syllabus and textbook and not load all children with homework requiring time at home, often with parental/adult support. CCE requires teachers to create tasks for each child, but instead standard periodic tests are given to all. This conceptualization and implementation of CCE is hostile to its principles.

The urge for centralized control, disrespect for and lack of faith in teachers and disdain for the entire decentralized machinery, has forced CCE into becoming just another excuse for centralized testing. The teachers, already stressed by their inability to deal with diverse children, have failed to grasp the essence of CCE. Moreover, the RTE 2009 itself misrepresented CCE by not entirely dissolving grades and classes and discarding words like detention and promotion. Making an external body like the SCERT (or other academic authority) responsible for defining and monitoring indicates a widely held suspicion of the teacher.

The real issue is the absence of patience and a disinclination for sustained investment for equitable inclusive education. Financing universal education of the poor is seen as a drag on the economy, even as investment on middle class and rich children, subsidized through tax rebates on children's education, is not considered as either excessive or undesirable. Clearly, it is only in the context of education of poor children that we worry about the salary of teachers.

The much-lauded increased expenditure on primary and elementary education in the '90s and early 2000s was focused at infrastructure. There was no money available for teachers and the pitiful allocation for training was structured in a way that it became meaningless. In-service teacher training was so shabby that it alienated teachers by making them feel ignorant and incompetent. As a result, teachers were demotivated and many of them turned cynical. Even this increased outlay has now been rolled back and replaced by a lack of conviction that the system can function. However, the current clamour for testing, measurement and monitoring is misplaced since we do not want to ensure what is actually needed – expenditure on universal quality access with equity. While occasional testing and measurement at all levels may be of use if designed with the participation of those involved in schooling, the current administratively directed centralized assessment and 'knee-jerk' measures are a sure recipe for further disaster.

Learning to realize education's promise

DEON FILMER and HALSEY ROGERS

SCHOOLING is not the same as learning. In Kenya, Tanzania, and Uganda, when grade 3 students were asked to read a simple sentence like, 'The name of the dog is Puppy', three-quarters did not understand what it said. In rural India, nearly three-quarters of students in grade 3 could not solve a two-digit subtraction such as 46-17 – and by grade 5 half could still not do so. Although the skills of Brazilian 15-year-olds have improved, if they continue to improve at their current rate they will not reach the rich country average score in math for 75 years. In reading, it will take 263 years. These are all countries that have measured learning and made the results public; in too many other countries the problem remains hidden.

Schooling without learning is not just a wasted opportunity, but also a great injustice: the children society is failing the most are the ones in greatest need of a good education to succeed in life. Without learning, education fails to deliver fully on its promise as a driver of poverty elimination and shared prosperity. Within countries, learning outcomes are almost always much worse for the disadvantaged. In Uruguay, poor children in grade 6 are assessed as 'not competent' in math at five times the rate of wealthy children. Moreover, these results are for children and youth lucky enough to be in school. Many aren't even enrolled in primary or secondary school, with members of disadvantaged groups – poor children, girls, children with disabilities, ethnic minorities – most likely to be out of school. Together, these severe shortfalls constitute a *learning crisis*.

Education should equip students with the skills they need to lead healthy, productive, meaningful lives. Different countries will define skills differently, but all share some core aspirations, embodied in their curriculums. Students everywhere must learn how to interpret many types of written passages – from medication labels to job offers, from bank statements to great literature. They have to understand how numbers work so that they can make transactions in markets, set family budgets, interpret loan agreements, or write engineering software. They require the higher order reasoning and creativity that builds on these foundational skills. And they need the socio-emotional skills – like perseverance and the ability to work on teams – that help them apply what they have learned. But the learning that one would expect to happen in schools – whether expectations are based on formal curriculums, needs of employers, or just common sense – is often not occurring, especially for the already marginalized.

Learning shortfalls during the school years eventually show up as weak skills in the workforce – so when people in countries around the world debate the job-skills problem, they're really talking about the learning crisis. Because education systems have not prepared workers adequately, many enter the labour force with inadequate skills. Recent initiatives have assessed a range of skills in the adult populations of numerous countries. They find that even foundational skills such as literacy and numeracy are often low, let alone more advanced skills. The problem is not just a lack of trained workers; it is a lack of readily trainable workers. Accordingly, many workers end up in jobs that require minimal amounts of reading or math. A lack of skills reduces job quality, earnings and labour mobility.

The good news is that there is nothing inevitable about low learning in low and middle income countries: when improving learning is a priority, great progress is possible. Starting in the early 1950s as a war-torn society with very low literacy rates, the Republic of Korea had by 1995 achieved universal enrolment in high quality education through secondary school. In fact, its young people were performing at the highest levels on international learning assessments. Vietnam surprised the world when the 2012 results of the Programme for International Student Assessment (PISA) showed that its 15-year-olds were performing at the same level as Germany's, despite Vietnam's much lower income. Between 2009 and 2015, Peru achieved some of the fastest growth in overall learning outcomes due to concerted policy action and system reform. And in Liberia, Papua New Guinea, and Tonga, early grade reading improved substantially within a short time thanks to focused evidence based efforts.

But tackling the learning crisis requires first understanding what causes it. Around the world, the crisis has three main dimensions. The first is the poor learning outcomes themselves. As described earlier, *levels of learning are low*, and not just in the poorest countries. Many high performing students in some middle income countries (such as Algeria, the Dominican Republic, or Kosovo) would rank in the bottom quarter of students in the average Organization for Economic Co-operation and Development (OECD) country.

At the same time, *inequalities in learning outcomes are high.* By the end of primary school, only 5% of girls in Cameroon from the poorest quintile of households have learned enough to continue school, compared with 76% of girls from the richest quintile. And *improvements in system-wide learning are often slow*. In fact, across all countries participating in multiple rounds of the PISA since 2003, the median gain in the national average score from one round to the next was zero.

The second dimension of the learning crisis is its immediate causes, visible in schools in the various ways in which the teaching-learning relationship breaks down. The *World Development Report 2018* (WDR 2018) identifies four key ways this happens:

1. Because of poverty, many children arrive at school unprepared to learn. Malnutrition, illness, low parental investments, and the harsh environments associated with poverty undermine early childhood learning. Thirty per cent of children under 5 in developing countries are physically stunted – that is, they have low height for their age, typically due to chronic malnutrition. Poor developmental foundations mean that many children arrive at school unprepared to benefit fully from it, with poor children's cognitive skills falling well behind in the years before primary school.

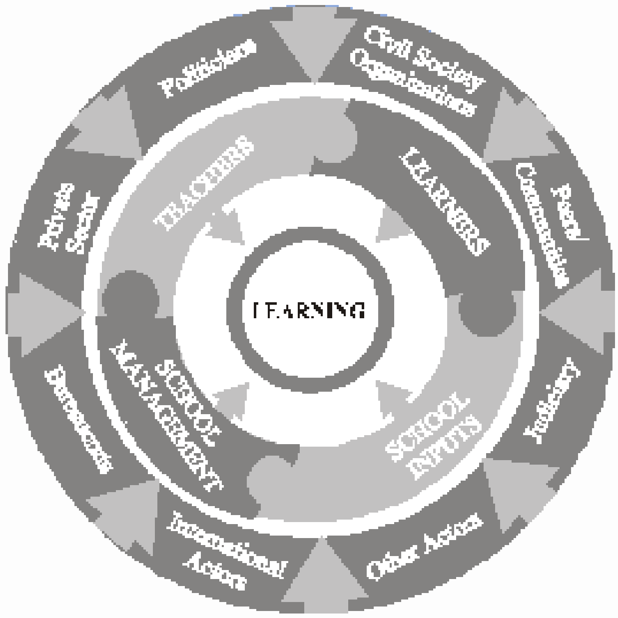
In some countries, the gap between richer and poorer children's ability to recognize letters of the alphabet doubles between the ages of 3 and 5. Moreover, the fees and opportunity costs of schooling keep many young people out of school, and the social dimensions of exclusion – for example, barriers linked to gender or disability – exacerbate the problem. These inequalities in school participation widen the gaps in learning outcomes.

2. Teachers often lack the skills or motivation to teach effectively. Teachers are the most important factor affecting learning in schools. In the United States, students with great teachers learn three times as fast as those with poor teachers; in developing countries, teacher quality can matter even more. But most education systems do not attract applicants with strong backgrounds or train teachers effectively. For example, 15-year-old students who aspire to be teachers score below average on the PISA in nearly all countries. Across 14 Sub-Saharan countries, the average grade 6 teacher performs no better on reading tests than the highest performing grade 6 students. Moreover, much learning time is lost because classroom time is spent on other activities or teachers are absent. In seven Sub-Saharan countries, one in five teachers was absent from school during recent

unannounced visits by survey teams, with another fifth of teachers at school but absent from the classroom. These diagnostics are not intended to blame teachers; rather, they call attention to how education systems undermine learning by failing to support their frontline educators.

3. Inputs often fail to reach classrooms or to affect learning. Public discourse often equates problems of education quality with input gaps. Devoting enough resources to education is crucial, but input shortages explain only a small part of the learning crisis. One reason is that inputs often fail to make it to the frontlines. In Sierra Leone, for example, textbooks were distributed to schools, but follow up inspections found most of them locked away in cupboards, unused. Similarly, many technological interventions fail before they reach classrooms, and even when they do make it to classrooms, often do not enhance teaching or learning. In Brazil, the One Laptop Per Child initiative in several states faced years of delays. Then, even a year after the laptops finally made it to classrooms, more than 40% of teachers reported never or rarely using them in classroom activities.

Coherence and Alignment Toward Learning



Source: WDR 2018 team.

4. Poor management and governance often undermine schooling quality. Although effective school leadership does not improve student learning directly, it does so indirectly by strengthening teaching and ensuring effective use of resources. Careful analysis of management practices shows that schools in developing countries often suffer from poor management, compared with schools in richer countries or even with manufacturing firms in their own countries. Ineffective school leadership means that school principals are not actively involved in helping teachers solve problems, do not provide instructional advice, and do not set goals that prioritize learning. Moreover, in many settings schools lack any meaningful autonomy, and community engagement fails to affect what happens in classrooms.

The third dimension of the crisis is its deeper systemic causes. All these breakdowns in schools and communities are driven by deeper system level factors – often invisible – that pull key actors away from a focus on learning. For one thing, operating an education system effectively poses major *technical* challenges: parts of the system need to be aligned with learning and coherent with each other, and actors at all levels need the capacity to implement well. But many of the deeper causes of the learning crisis are *political* in nature. Actors have interests that diverge from learning. Politicians act to preserve their positions of power, which may lead them to target certain groups (geographic, ethnic, or economic) for benefits. Bureaucrats may focus more on keeping politicians and teachers happy than on promoting student learning, or they may simply try to protect their own positions.

Private suppliers of education services – whether textbooks, construction, or schooling – may, in the pursuit of profit, advocate policy choices that undermine learning. Teachers and other education professionals, even when motivated by a sense of mission, also may fight to maintain secure employment and to protect their incomes. None of this is to say that education actors don't care about learning – rather, especially in poorly managed systems, competing interests may loom larger than the learning-aligned interests. These technical and political challenges keep many systems stuck in low-learning traps, with low accountability and high inequality.

To do better, a nation must assess learning, to make it a serious goal; act on evidence, to make schools work for learners, and align actors, to make the system work for learning.

First, assess learning, to make it a serious goal. Only half of all countries have metrics to measure learning at the end of primary and lower secondary school – indicators that are required to monitor progress toward the United Nations' Sustainable Development Goals for learning. Fewer still have the ability to track learning over time.

Countries need to put in place a range of well designed student assessments to help teachers guide students, improve system management, and focus society's attention on learning. These measures can put the spotlight on hidden exclusions, inform policy choices, and track progress.

Second, act on evidence, to make schools work for learners. Great schools are those that build strong teaching-learning relationships in the classroom. Thanks to innovations by educators and advances in brain science, knowledge of how students learn most effectively has exploded. But common practice in schools and communities often diverges sharply from what evidence identifies as most promising.

Countries can use the evidence to close this gap and make schools work better. The best place to start is in these three key areas.

Prepared learners: Reduce stunting and promote brain development through early nutrition and stimulation (as in Chile) so children can learn. Support disadvantaged children with grants to keep them in school.

Skilled, motivated teachers: Attract talented people into teaching (as in Finland). Use repeated, specific teacher training reinforced by mentors (as in some African settings) instead of the ineffective one-off methods that are more common.

Inputs and management focused on teaching and learning: Deploy technologies that help teachers teach to the level of the student (as in Delhi). Strengthen the capacity and powers of school management (as in Indonesia), including principals.

Third, align actors, to make the whole system work for learning. All this innovation in classrooms is not likely to have much of an impact if system level technical and political factors prevent an emphasis on learning. When key actors are focused on non-learning goals (such as political or personal gain) or lack implementation capacity, even well designed innovations can't be scaled up or sustained.

Countries can escape low-learning traps by acting on three fronts as they implement reforms: (*i*) Deploy information and metrics to make learning politically salient (as the NGO-led ASER and Uwezo programmes have done in India and East Africa); (*ii*) Build coalitions to shift political incentives toward learning for all (as Chile did early in its decades-long education reforms, or as Malaysia and Tanzania did recently with their collaborative societywide ‘labs’ to design reform programmes); and (*iii*) Use innovative and adaptive approaches to find out which approaches work best in their context (as Burundi did while rebuilding its education sector after conflict).

Together, these three policy actions can deliver a system in which the elements are coherent with each other and all elements align with learning (see figure).

Coherence and alignment toward learning are necessary to ensure that a society’s investments in education pay-off fully. Given the many returns to education – financial and nonfinancial, for both individuals and societies – some countries clearly need to invest more in education, especially as more youth complete primary and secondary school and go on to higher education. At the same time, virtually all countries should spend more effectively. Across countries, the relationship between public spending on education and learning outcomes is weak, but the right investments can pay off. Used in conjunction with the actions outlined here, financing for education can help pull systems out of low-learning traps and expand opportunity.

The pay-off to these efforts is education that delivers for growth and development. Delivered well, education cures a host of societal ills. For individuals, it promotes employment, earnings, health and poverty reduction. For societies, it drives long-term economic growth, spurs innovation, strengthens institutions and fosters social cohesion. But as the mounting evidence shows, it is the *skills* acquired through education, rather than just the years spent in school, that drive growth and equip individuals for work and life.

Reforms to strengthen learning will allow countries to reap the many benefits of education. For example, even a relatively modest improvement in learning – one that lifts all students to the level of the average student in Brazil – could increase long-term growth rates in a middle income country like Mexico or Turkey by around two percentage points. Rapid technological change makes these foundational skills even more important because it is those skills that enable workers and citizens to adapt rapidly to new opportunities. Countries have already made a start by getting so many children and youth into school. Now it is time to realize education’s promise by accelerating learning for all.

* This article is drawn from the *World Development Report 2018: Learning to Realize Education s Promise*. Citations for the figures and assertions in the article can be found in the report, which is available at www.world bank. org/wdr2018.